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Review

Interpretable intelligent intrusion detection systems for cybersecurity in smart digital substations

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ABSTRACT

Cybersecurity in Smart Digital Substations (SDS) is becoming more challenging in recent years due to the integration of Information and Communication Infrastructure (ICT), which makes this critical infrastructure prone to cybersecurity threats. Existing Intrusion Detection Systems (IDS) rely on black-box Machine Learning (ML) and Deep Learning (DL) based Artificial Intelligence (AI) models, which lack explainability and transparency, and thus the detection decisions of these IDS are not trusted. Explainable Artificial Intelligence (XAI) techniques seek to make these previously untrustworthy black-box detection schemes in SDS explainable. However, the implementation of XAI-based IDS in SDS is not trivial as it brings many challenges regarding which XAI technique is suitable as well as the selection of XAI parameters, impacting the interpretability, stability, and faithfulness of the explanations. To the best knowledge of the authors, this work is the first experimental review that investigates several XAI techniques in terms of their parameters, human interpretability, stability, faithfulness, and fairness in the context of SDS cybersecurity using a realistic IEC 61850 testbed and a parameter-level sensitivity analysis to provide practical analysis. This work aims to guide researchers and users in the appropriate selection and implementation of XAI techniques in future ML-based and DL-based IDS in SDS, ensuring trust and secure use.

1. Introduction

1.1. Problem statement and motivation

Smart Digital Substations (SDS) rely heavily on Information and Communication Technology (ICT), which make such critical infrastructure a target for cybersecurity threats. Intrusion Detection Systems (IDS) aim to detect such cyber-attacks to prevent failure of substation devices and prolonged power outages [1]. Several existing IDS are typically based on Machine Learning (ML) and Deep Learning (DL) Artificial Intelligence (AI) models, which act as black-boxes and thus the detection decisions lack explainability and trust. SDS themselves are systems within the modern Smart Grid (SG), which are a subset of broader Cyber-Physical Systems (CPS).

With the widespread adoption of black-box models across industries, including the SDS within the SG, legislation has begun to be introduced that requires decisions taken by models to be understandable [2]. The lack of interpretability of such models has been a focus of the European Union's General Data Protection Regulation [1], which requires that the decision made by an ML algorithm that can negatively impact individuals to be explainable. In safety-critical applications like SG and SDS, the lack of explainability of the models limits trust, regulatory compliance, and real-world applicability [3]. In an SDS, without explanations of how an AI model came to a conclusion, operators are not likely to take actions as if an incorrect action is taken significant costs can be incurred as critical equipment is put at risk and service to connected customers can be disrupted. To address the lack of explainability, Explainable Artificial Intelligence (XAI) can be applied [4], allowing a cybersecurity system to defend its decisions with justifications. This is critical in a SDS context where the output of the model may be used for protection decisions that impact critical equipment and potentially lives of employees or customers. With such high stakes, SDS operators need to understand how an AI model made its decision to verify that it's working correctly and can then trust it, which XAI facilitates. However, the implementation of XAI-based IDSs still faces several challenges due to different explanation capabilities and complexities of existing XAI techniques.

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1.2. Previous work

In the context of SDS a cyberattack detection scheme must be trustworthy as a false action can incur significant costs. Overall, many existing IDS are not explainable [5]. Specifically, in SDS, only the work in [6] applies XAI to detect attacks but does not offer a deep investigation of what XAI methodology was used and why, and how the parameters of the methodology were selected. There is a lack of work on an analysis of the XAI techniques that apply cybersecurity in SDS. Within the broader SG, and furthermore CPS, XAI has been investigated in a limited fashion with the review works of [3–8], which are compared in Table 1. It is notable that only [3] and [4] review XAI from a SG perspective, and [5–8] from a CPS perspective, but none from a SDS perspective. Additionally, the review works report applications of and research conducted on XAI, but not what XAI techniques are available, how to apply them, or the quality of the explanations. Additionally, no work in this field explains the parameters of the XAI techniques and their effect on the generated explanations using sensitivity analyses. These parameters turn out to significantly affect explanations and are critical. As such, no work provides guidance on how to select the parameters of an XAI technique to improve explanations. What XAI technique to use, and how to use it effectively, is not fully investigated in the literature.

From this analysis of the literature the following research gaps have been identified:

- G1: A lack of detailed information on what XAI techniques are available and their uses in SDS cybersecurity.
- G2: A lack of investigation of the strengths and weaknesses of XAI techniques in a SDS cybersecurity context.
- G3: A lack of analysis of the parameters of XAI techniques and their impacts on explanations.
- G4: A lack of guidance regarding the selection of XAI techniques.

This work seeks to bridge these gaps through performing an experimental review of XAI methodologies, their explanation's capabilities, interpretability, complexity, and their parameters from a practical perspective using realistic experimental data gathered from a SDS test system.

1.3. Contributions

This work is an experimental review that contains contributions C1 through C3. These contributions map to the research gaps as contribution C1 addresses G1 and G2, while C2 addresses G1 and G3, and contribution C3 addresses G4.

- C1: An in-depth experimental review of the implementation of XAI-based IDS in SDS with respect to the human interpretability, stability, faithfulness, and fairness in a realistic application.
- C2: The experimental-based review of the capability of existing XAI techniques and the effect of their parameter selection using a sensitivity analysis on the performance of an XAI-based IDS in the context of SDS cybersecurity using an experimental dataset representative of a SDS cybersecurity scenario.
- C3: Guidance on the appropriate selection and implementation of XAI techniques in future AI-based IDS to improve explanations to ensure their trust and secure use in SDS.

Table 1
Relevant works reviewing XAI in the literature in CPS.

Reference	SDS Focus?	XAI Technique			
		Explanation Quality Investigation	Parameter Identification	Parameter Sensitivity Analysis	Guidance
[3]	×	×	×	×	×
[4]	×	×	×	×	×
[5]	×	×	×	×	×
[7]	×	×	×	×	×
[8]	×	×	×	×	×
This Work	✓	✓	✓	✓	✓

2. Explainable artificial intelligence techniques

Existing XAI techniques used in the broader SG context include SHapely-Additive exPlanations (SHAP) [9–18], Local Interpretable Model-agnostic Explanations (LIME) [9–11,18,19], and Occlusion Sensitivity Mapping (OSM) in [6]. These methods are summarized in Table 2. Regarding experimental works, [6] utilizes OSM to explain the decisions of an IEC 61850-based detection scheme, while the work in [9] uses both SHAP and LIME to explain photovoltaic (PV) power generation forecasting model results. However, these works lack any analysis of the parameters of the XAI methods used and do not compare them thoroughly in terms of the quality of their explanations.

In [10], both SHAP and LIME was used as well to investigate PV power forecasting models, but differences in the explanations between the two methodologies are identified but not explained or investigated. Similarly, Akpuluma et al. [11] apply SHAP and LIME as well to PV power generation forecasting but only present XAI as a step to acquire explanations of the model and does not investigate the XAI methodologies. Notably, the work takes the XAI explanations at face value and does not investigate the meaningfulness of the outputs. Meanwhile, in [12] and [13] SHAP is used to explain key features again in power generation forecasting, but both works only identify SHAP as the XAI method used and use the explanations from SHAP with no validation. The work in [14] utilizes SHAP explanations within an adversarial training-based anomaly detector for false data injection in a SG context but lacks any investigation into SHAP and the quality of the explanations used. In this context better explanations could lead to improved detection accuracies.

The work in [15] utilizes SHAP explanations in a SG anomaly detector. However, no investigation is performed into the meaningfulness of the explanations, and as the detector relies on SHAP explanations, improvements in the quality of the explanations can improve the detection accuracy, or conversely, poor explanations can worsen the accuracy. In this scenario, informed selection of the XAI methodology and its settings is key but is lacking. The works in [16] and [17] both propose XAI to identify the importance of features for a SG IDS, with [16] using just SHAP, and [17] using both SHAP and LIME. However, both works take the explanations at face value and do not further analyze the parameters used, along with [17] failing to identify or investigate the differences in explanations between the implementations of SHAP and LIME.

Both SHAP and LIME were also used in [18] to explain a Supervisory Control and Data Acquisition (SCADA) IDS and some analysis was provided into the results of the methods, identifying LIME as being more useful in the given scenario. However, no attention was paid to the parameters of the XAI methods, and other XAI methods are not discussed. LIME is used in [19] to explain a PV power generation forecast model, but also with no analysis of the parameters of LIME used, or the impact of the choice of XAI model. There is a lack of a complete parameter-level sensitivity analysis across all the existing works. These existing experimental works apply XAI, but do not consider how it is applied or the quality of the explanations. These works could benefit from a work that identifies the XAI methodologies and provides guidance on how to use and setup the XAI methodologies.

To expand the analysis with additional XAI techniques that have not necessarily been applied in a SDS cybersecurity context yet, model-agnostic methods investigated in this work include SHAP [20], LIME [21], OSM [22], and Randomized Input Sampling for Explanation (RISE) [3]. These methods can be applied independently of the AI model chosen. Additionally, this work includes the non-model-agnostic methods

Table 2
Relevant experimental works that apply XAI in the SG.

Reference	XAI Techniques Considered				Investigation of Impact of Parameter Selection	Parameter-Level Sensitivity Analysis	Investigation of Impact of XAI Method Selection
	OSM	LIME	SHAP	Other Methods Listed in This Work			
[6]	✓	×	×	×	×	×	×
[9]	×	✓	✓	×	×	×	×
[10]	×	✓	✓	×	×	×	×
[11]	×	✓	✓	×	×	×	×
[12]	×	×	✓	×	×	×	×
[13]	×	×	✓	×	×	×	×
[14]	×	×	✓	×	×	×	×
[15]	×	×	✓	×	×	×	×
[16]	×	×	✓	×	×	×	×
[17]	×	✓	✓	×	×	×	×
[18]	×	✓	✓	×	×	×	✓
[19]	×	✓	×	×	×	×	×
This Work	✓	✓	✓	✓	✓	✓	✓

such as: Vanilla Gradient (VG) [24], SmoothGrad [25], and Gradient-weighted Class Activation Mapping (Grad-CAM) [26]. These non-model-agnostic methods take advantage of the structure of a Convolutional Neural Network (CNN).

2.1. Vanilla gradient (VG)

VG generates a saliency map for a CNN by first passing the input image for a classification forward through a CNN to classify the image and then determines the gradient of the class score with respect to the classification based on the input pixels with backpropagation [24]. The resulting saliency map is the gradients with respect to the input pixels.

2.2. SmoothGrad

Introduced in [25], SmoothGrad extends VG. It adds Gaussian noise to the input data with a variance of σ^2 and then classifies the case and performs backpropagation to determine the VG saliency map. This is then repeated for $N_{smthgrd}$ samples. The resulting $N_{smthgrd}$ maps are then averaged to return a single saliency map.

2.3. Gradient-weighted class activation mapping (Grad-CAM)

Grad-CAM is a method applied to CNNs that generates Grad-CAM saliency maps [26]. Grad-CAM saliency maps are calculated starting from the vector of the scores y_c for every class c. Next, the gradient of this vector is calculated against the feature map activations of the convolution layers, A^k, through backpropagation. The gradients are then global-average-pooled over the dimensions of the layer, x and y, to yield the neuron importance weights α_k^c, in (1), where N is the number of pixels in the feature map.

$$\alpha_k^c = \frac{1}{N}\sum_x\sum_y \frac{\partial y^c}{\partial A_{x,y}^k} \tag{1}$$

The weighted combination of the feature map activations with the importance weights is passed through a Rectified Linear Unit (ReLU) layer to yield the Grad-CAM saliency map, $M_{Grad-CAM}^c$:

$$M_{Grad-CAM}^c = ReLU\left(\sum_k \alpha_k^c A^k\right) \tag{2}$$

2.4. SHapely-Additive exPlanations (SHAP)

SHAP is a model-agnostic XAI methodology measures the contribution of each feature to the classification of an ML model based on Shapely values [24]. The explanation of a model is given as a linear addition of the importances of each feature [20]. In this work, Kernel SHAP is utilized which samples vectors representing the presence of features in the input data, after which the chosen vectors are classified and the weight for the coalition vector is computed. A linear model is fit whose coefficients represent the Shapely values of the features [24]. Kernel SHAP weights each coalition based on the number of features it includes, effectively estimating the Shapely values, which indicate the importance of each feature. SHAP suffers from high computational complexity as the number of features increase so segmentation is used to reduce images into a smaller set of representative features. In this work the SHAP plots are depicted in the traditional blue/red colour scheme, but for the calculations the resulting SHAP explanations are formatted and scaled just as the explanations of the other XAI methods investigated.

2.5. Local interpretable model-agnostic explanations (LIME)

Proposed in [21], LIME is an XAI tool that is model-agnostic. It builds a simple interpretable model around a test case and returns feature importance maps. The interpretable model is trained based on samples of the original model such that the model is faithful to the behaviour of the original more complex model among the sampled regions [21]. In the context of image-based classification with LIME, images are broken down into a smaller set of features using segmentation. Synthetic samples are created by randomly excluding features from the image through occlusion and are then classified using the large model. The synthetic samples and their classifications are then used to train a simpler model. From the simpler model the importances of the features are determined and mapped back to the segmented image to generate feature importance maps [27].

2.6. Occlusion sensitivity mapping (OSM)

Proposed in [22], OSM is model-agnostic and generates maps of the change in total activation of the model as in Fig. 1. This method systematically occludes portions of an input image and then classifies the occluded image using the model in question. An occluding patch is slid across the input image and the change in the total activation of the model is plotted against the position of the occluding mask, creating a map of the change in total activation [27].

2.7. Randomized input sampling for explanation (RISE)

RISE is a model-agnostic XAI method proposed in [23] that generates maps of pixel importances. Randomized occlusion masks are applied to the input image, and a weighted sum of the classification score output of the desired class and the occlusion mask is gathered. The result is a map of pixel importances, the RISE map. In [23] the mask intensities are considered to completely affect the occluded pixels by setting them to

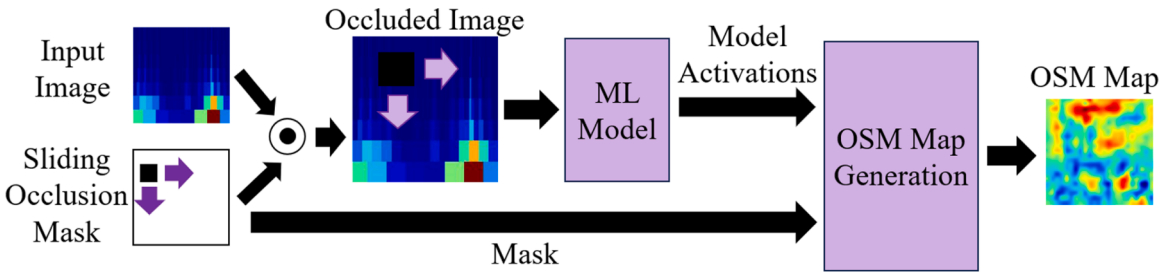


Fig. 1. A graphical breakdown of OSM showing the use of the sliding occlusion mask.

zero. The masks are generated by first generating a matrix of zeros and ones of a smaller size than the input image and then upscaling the matrix to the input image size using bicubic interpolation to provide soft transitions. The masks are then randomly translated vertically and horizontally to ensure that the edges of the masks do not affect the features. These masks are multiplied with the input image to result in masked images, which are classified using the model to yield classification scores, which are used as weights to sum the masks to yield the final RISE map, shown in Fig. 2.

3. XAI-Based intrusion detection system for SDS

Fig. 3 depicts the proposed XAI-Based IDS for SDS, which will be used to evaluate the XAI techniques that can be integrated with an IDS for SDS. The system is an IDS-based on [28] but appended with an XAI Explainability Module. The IDS proposed in [28] is chosen for cyber-attack detection as it focuses specifically on SDS and is representative of an ML approach. This methodology centers on converting the collected data into time-frequency spectrograms to extract time-frequency features that improve the accuracy of the IDS [28]. Cyber and physical data can be used for detection with this methodology, offering detection capabilities across a wide variety of systems. This comes at the cost of additional preprocessing complexity and the need for image-based classification methods.

In the case of this work, the additional classification accuracy performance was desired with the trade off of simple processing and a simpler detection model. This Explainability Module adds trustworthiness to the IDS-based system in [28] as it applies XAI to generate image-based explanations of the decisions of the IDS. The breakdown of the IDS in Fig. 3 involves preprocessing data collected from a SDS, extracting time-frequency features into spectrograms using the Dual-Tree Complex Wavelet Transform (DT-CWT), and classifying the spectrograms using the existing pretrained CNN AlexNet. The novel Explainability Module, added in this work, explains the decisions of the CNN, adding trustworthiness to the explainability of the decisions.

The test system and dataset from [28] represent a SDS that uses IEC 61850 Generic Object-Oriented Substation Event (GOOSE) communication that contains several attack types. The original system, from [29], was extended with new attack types and additional cases. The expanded system includes False Data Injection (FDI), Replay, Message Suppression (MS), Denial-of-Service (DoS), and Insider attacks alongside additional power quality disturbance scenarios. A visual summary of the test system used to generate the dataset is provided in Fig. 4. Five virtual Intelligent Electronic Devices (IED) monitor a simulated substation and communicate with each other over a network using real GOOSE messages, allowed for by an OPAL-RT OP5033XG real-time digital simulator. The reader may refer to [28] for an in-depth explanation of the design of the system. The GOOSE features dissected from the collected packets are described in [28], alongside a windowing methodology used to generate the final dataset, which is outlined in Table 3. The dataset used mainly impacts the training and testing of the classifier, and not the XAI methods, hence a dataset representative of the test system, as is used here achieves the goal of this work. This dataset is imbalanced, as is likely to occur real-world applications, and allows for a testing of the performance of the XAI methods in terms of fairness when imbalanced classes are present.

A test case for analysis in this work is taken from the dataset generated in [28] and is of a DoS attack test case. The input data contains features collected from IEC 61850 GOOSE traffic from the test system, including physical measurements of currents and voltages, as

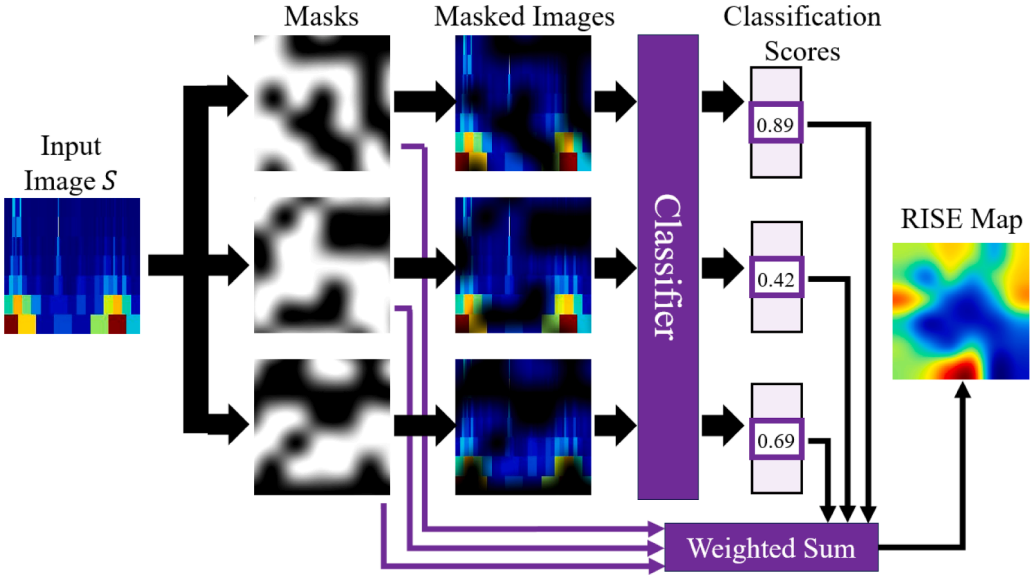


Fig. 2. A graphical breakdown of RISE highlighting the use of occlusion masks and a weighted sum to generate the RISE map.

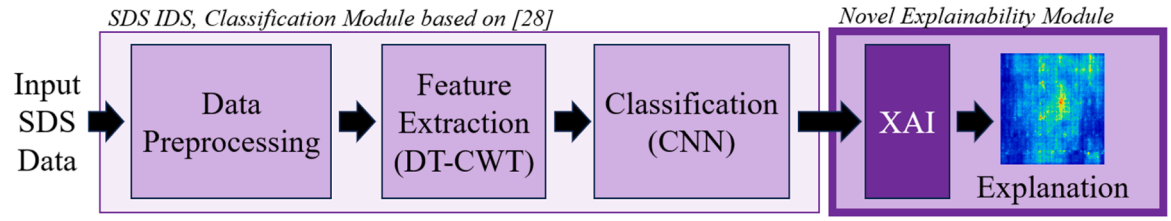


Fig. 3. An overview of the proposed XAI-based IDS for SDS, with the new Explainability Module, outlined in purple, appended to the existing SDS IDS methodology proposed in [28].

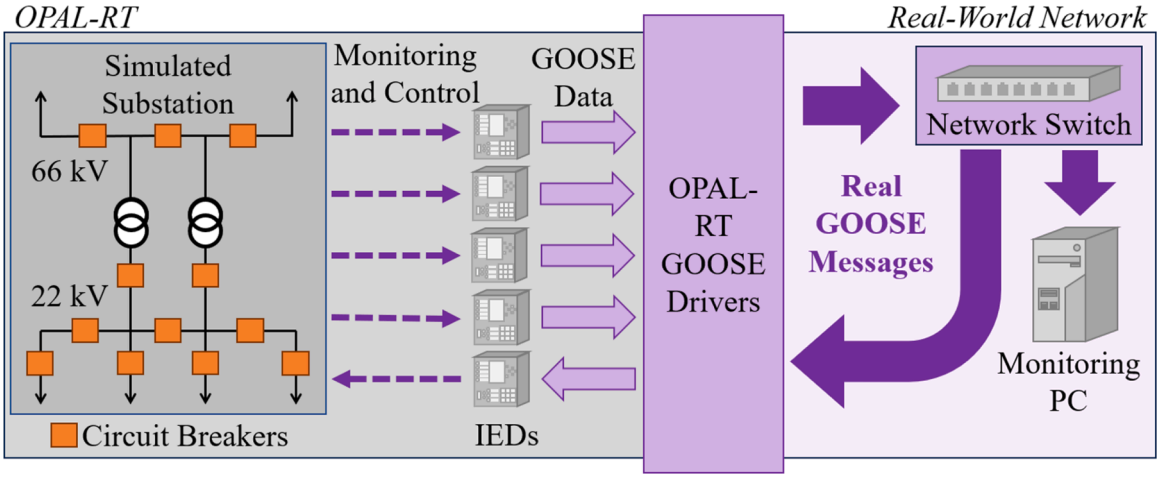


Fig. 4. An overview of the SDS test system used in this work, based on a simulated substation deployed on a real-time simulator that communicates over a real-world network using real GOOSE messages.

Table 3
A breakdown of the testing dataset used in this work.

Class	Number of Cases
Normal	79
Disturbance	22
Attack, FDI	83
Attack, Replay	12
Attack, MS	36
Attack, DoS	100
Attack, Insider	66
Total Cases	**398**

well as features related to the GOOSE messages themselves. This test case is correctly classified by the trained CNN classifier from [28]. XAI methodologies are applied to explain the why the CNN made its choice. The spectrogram of the test case for classification is in Fig. 5(a), and an example saliency map explaining the decision from this case in Fig. 5(b), showing how the CNN focused on the feature in the bottom corner of the spectrogram. The area of highest magnitude in the saliency map, outlined in red, was most important for classification, and is linked to the feature highlighted in red in the spectrogram, which lies in the same region. Regarding other attack type examples and how the explanations are related to the spectrograms, additional test cases are provided and described in Fig. 5. Regarding the FDI test case, the explanation highlights the region outlined in magenta in Fig. 5d), corresponding to a strong feature in the same region of the spectrogram in Fig. 5c). In the Replay attack case, the explanation is concentrated in the region outlined in magenta in Fig. 5f), which is a distinct feature in the same region in the spectrogram in Fig. 5e) outlined similarly. For the MS attack case the explanation focuses on two features, outlined in magenta and orange in Fig. 5h). In the spectrogram in Fig. 5g) the magenta region corresponds to the lack of features of that portion of the spectrogram, while the orange region contains a strong feature. In this case the decision was made due to these two areas in conjunction. Lastly, in the Insider attack case the explanation in Fig. 5j) also has two main features, a large feature outlined in magenta and a small feature outlined in orange. In the spectrogram for this case in Fig. 5i), the large feature again corresponds to a mostly empty region of the spectrogram outlined in magenta, while the smaller feature focused on a weak but unique feature, outlined in orange. Across the attack types the CNN learned different features.

A CNN is used as a representative model as in the literature this is a well-established classification method and offers several CNN-specific XAI methods. For other types of AI models, the model-agnostic methodologies identified in this work can be applied. Since the test system consists of the classification of spectrograms as images, XAI methodologies that are applicable to images are investigated in this work. Data not in an image form can be converted into time-frequency spectrograms using time-frequency transforms like wavelet transforms, and then XAI techniques are applied.

4. XAI Techniques parameter analysis

In this section the parameters of the identified XAI techniques are investigated experimentally. The techniques and parameters that were analyzed are summarized in Table 4. Each technique is evaluated in terms of the identified parameters, and the key issue of each method is discussed. An explanation of the quantitative metrics used to analyze the methods is first provided.

4.1. Parameter evaluation technique

The investigation in this work serves as a sensitivity analysis of each parameter. A standard One-at-a-Time (OAT) sensitivity analysis approach is used so that each parameter individually could be

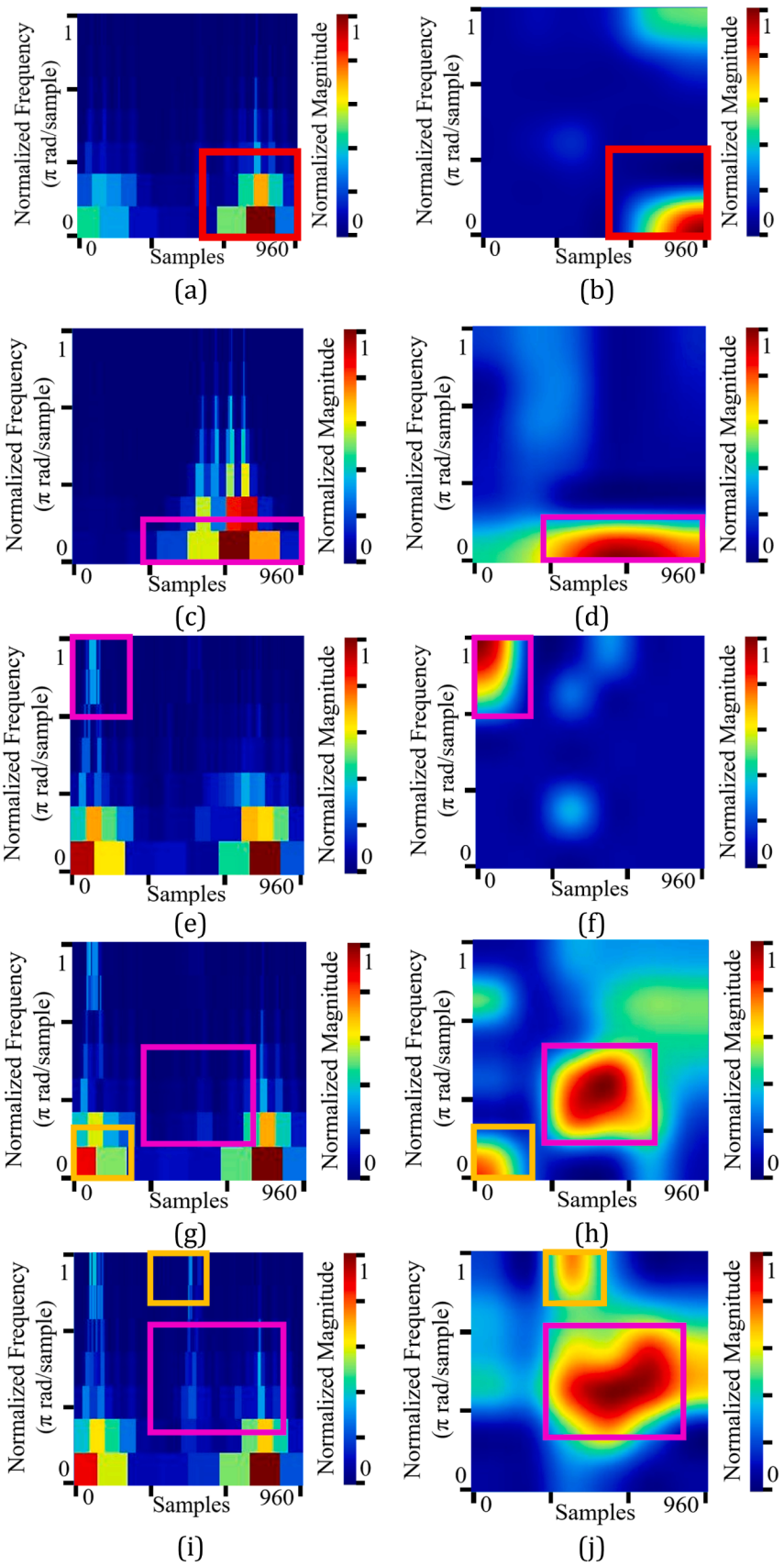


Fig. 5. The generated time-frequency spectrogram and explanation of the proposed XAI-based IDS in the case of a DoS, FDI, Replay, MS, and Insider attacks. The corresponding time-frequency spectrograms are in (a) (c) (e) (g) and (i), while the corresponding saliency maps generated using Grad-CAM are shown in (b) (d) (f) (h) and (j).

Table 4
A summary of the XAI techniques and the parameters that are investigated.

XAI Technique	Parameters Analyzed
VG	Gradient Representation
SmoothGrad	Gaussian Noise Variance
	Number of Noise Samples
	Gradient Representation
Grad-CAM	Feature Layer Selection
	Upsampling Method
SHAP	Image Segmentation Method
LIME	Number of Features
	Segmentation Method
	Number of Synthetic Images
	Model Type
OSM	Stride and Mask Size
	Mask Clipping
	Mask Replacement Value
RISE	Generated Mask Size
	Percentage of Input Image to Occlude
	Number of Samples

investigated. By understanding the individual impact of each parameter, a meaningful combination of parameters can be achieved, which is critical as useful explanations are desired. Additionally, the XAI parameters analyzed in this work independently affect the explanations, and as such OAT is applicable. Also, since the impacts of the parameters are not evaluated only using numerical metrics but also visually OAT was used. If more than one parameter is varied at a time visual analysis becomes more difficult and less meaningful.

4.2. Explanation quantitative comparison metrics

To better analyze the generated explanations from the XAI methodology, quantitative metrics are used to quantitatively compare the explanations. The interpretability, stability, faithfulness, and fairness of the XAI techniques are measured using the metrics summarized in Table 5. Interpretability is measured using sparsity and Total Variance (TV). Sparsity is measured using the Hoyer sparsity from [30] as it satisfies most desired sparse criteria as opposed to simpler sparsity measures. TV is from [31] and is a standard metric used in image processing. The stability metric was chosen to be from the perturbation-based and well-referenced work in [32]. The faithfulness metric measurement using deletion plots was used as it was for explanation evaluation in [23], and the Area Under the Curve (AUC) metric is derived directly from the plots. Fairness is measured by evaluating the equality of the resulting metrics on subgroups using various metrics as mentioned in classification ML fairness in [33]. The following sections describe the metrics in more detail.

4.2.1. Explanation human interpretability

One aspect of the quality of explanations is the ease of interpretability by a human. This human interpretability is measured using sparsity, and TV. These metrics evaluate visual structure and are interpreted in terms of their relation to true human interpretability but are themselves not direct measures of true human interpretability. As true human interpretability is subjective and qualitative, these metrics provide a quantitative ground for comparison. Regarding sparsity, Hoyer sparsity, used in [30], is given by (3), where E is an explanation, and n_p is the number of pixels of the explanation. This is a measure of how concentrated features are within the image, with a low value of 0 being a flat image, and a value approaching 1 being very concentrated, signifying overfitting.

$$S[E] = \frac{\sqrt{n} - \frac{\|E\|_1}{\|E\|_2}}{\sqrt{n} - 1} \tag{3}$$

TV, given in (4) is a measure of the spatial smoothness of the explanation [31]. In (4) i and j are indices, and H and W are the height and width of the explanation respectively. The discrete derivative is used and is normalized with the number of discrete derivates calculated. Low values indicate fewer visual variations, and high values indicate high levels of noise. It is notable that high frequency features are difficult for humans to interpret [34], so lower values are desired for human interpretability. It is also important to note that this metric is for relative comparisons.

$$TV[E] = \frac{1}{(H-1)(W-1)} \sum_{i=1}^{H-1} \sum_{j=1}^{W-1} \sqrt{\left(E_{i+1,j} - E_{i,j}\right)^2 + \left(E_{i,j+1} - E_{i,j}\right)^2} \tag{4}$$

4.2.2. Explanation stability

Regarding the robustness of the explanations, that is, the sensitivity of the explanations with respect to the input image, the Mean Squared Error (MSE) was calculated between the original explanation and an explanation of the same test case but with a small perturbation. This is taken in effect to be an indication of the stability of the output as it measures how much the explanation changes with a slight change in the input. This allows for the relative comparison of the stability of the XAI techniques. Since in this work all explanations are of the same resolution and kept to the same scale, MSE results in a fair comparison. To perturb the test case, Gaussian noise was added with a variance of 0.001 as this was found experimentally to be the maximum amount that could be added without causing the classifications of the model to change. This ensures that the perturbations remain local and maintain the behaviour of the CNN. The stability measure $\mathscr{S}$ is given by (5), where the explanation of the original test case is E, and Ep is the explanation of the perturbed test case. The summation is carried out over all elements.

$$\mathscr{S} = \sum_{i=1}^{H} \sum_{j=1}^{W} \left(E_{i,j} - Ep_{i,j}\right)^2 \tag{5}$$

To compare between two explanations to aid in the analysis of the impact of selection of parameters, the change between two explanations is measured using the MSE of the two images. The general MSE metric for two explanations, $E1$ and $E2$, is given by (6). This metric provides a way to measure stability with respect to the selected parameters for an XAI technique.

$$MSE = \sum_{i=1}^{H} \sum_{j=1}^{W} \left(E1_{i,j} - E2_{i,j}\right)^2 \tag{6}$$

4.2.3. Explanation trustworthiness/faithfulness

To address the trustworthiness of the explanations their faithfulness is measured using deletion plots, as in [23]. In this method each pixel of an explanation is first ranked in order of its magnitude. Incrementally the top p valued pixels have their corresponding locations in the input test case replaced with the mean value of the pixels of the input test case. In this work, p was varied from 0 to 0.25 in steps of 0.01, leading to up to 25% of the input image being deleted. Beyond 25% removal, random effects had more impact as the test case with deletions no longer was

Table 5
A summary of the quantitative comparisons provided and their evaluation metrics.

Qualitative Comparison	Evaluation Metrics
Interpretability	Sparsity
	Total Variance (TV)
Stability	Perturbations + MSE (Stability)
Faithfulness	Deletion Plots
	AUC
Fairness	Per-Class Sparsity
	Per-Class TV
	Per-Class Stability
	Geometric Mean of Per-Class Metrics
	Sample Variance of Per-Class Metrics

related to any of the test cases in the training data as it differed too much. For each value of p, the deleted input image was classified using the model and the resulting CNN's confidence value of the classification plotted. The result is a plot with a high confidence at p values close to zero, representing the initial image, with a rapid drop as p increases since the features key for classification are deleted. The initial drop in accuracy signifies the validity of the explanation to the model; that as the identified key regions are removed the confidence of the model drops significantly. To quantitatively compare methods, the AUC for the given range of p was calculated as a discrete integral. Lower AUC values indicate that the explanation is more faithful, allowing for a relative comparison between XAI techniques.

4.2.4. Explanation fairness

To analyze the fairness of the XAI techniques, how the explanations represent each class within the dataset was analyzed. First, the performance of the XAI technique on the testing dataset for each class is measured in terms of the mean of the per-class sparsity, TV, AUC, and stability metrics. Fairness is indicated by the equalness of these metrics across each class. If the metrics were to be identical for each class, the explanation would be the fairest. To measure the similarity first the Geometric Mean of each of the per-class metrics is calculated to provide a measure to see which classes deviated most from the mean. The Sample Variance for each metric is subsequently calculated. Lower Sample Variances indicate more fairness as they mean that the measured per-class metric does not vary between classes significantly. Additional analysis is provided by subtracting the Geometric Mean of each metric over all classes to measure the difference in each class by metric independently.

4.3. Vanilla gradient analysis

4.3.1. VG Parameter: gradient representation

In VG the representation of the gradients significantly affects the resulting saliency map. The maximum of the absolute value, minimum value, and maximum value of the gradient can be used, see Fig. 6. It is notable that both the regions of most interest, outlined in red, are in the same location for each run. The most useful information is acquired from using the absolute value of the weights, as can be seen in Fig. 6(a), it has more intense features than either of the other parameter options. Quantitatively, the interpretability metrics of each of the runs in terms of sparsity is 0.2442, 0.3913, and 0.3602, and in terms of TV is 0.0162, 0.0347, and 0.0721. From a human interpretability perspective, the maximum of the absolute value explanation has the lowest sparsity, indicating that it is flatter than the other cases. The resulting TV is also the lowest, indicating that it is the smoothest and most interpretable. This indicates that the data from negative gradients, that is pixels that negatively affect classification, and data from positive gradients, that is pixels that positively affect classification, are situated alongside each other making these features redundant visually. As a result VG only indicates regions of interest, and not whether areas positively or negatively help classification.

4.3.2. VG Key issue: variability

A key characteristic of VG is its variability. Due to the random dropout layers in the CNN, the weights within the network change significantly between runs using the same data. This is highlighted in Fig. 7, where VG is applied three times to the same DoS test case, but each time the resulting saliency map is visually different. In the first run, Fig. 7(a), a major feature is in the center of the spectrogram, outlined in red, while in the second run, Fig. 7(b), the map changes and new major features are seen in the upper right quadrant of the map, outlined in green. Furthermore, in the third run, Fig. 7(c), the major features change to the bottom of the map, outlined in orange. The major features visibly change significantly between runs. In terms of MSE, between the first and second run it is 0.0066, between the second and third run it is 0.0072, and between the first and third run it is 0.0052. This MSE is relatively high for explanations that should be identical given that the input data and test case has not changed, indicating that the explanations are changing significantly between runs. This variability greatly limits the usefulness of this technique as explanations are not stable.

4.4. SmoothGrad analysis

4.4.1. SmoothGrad parameter: Gaussian noise variance

In SmoothGrad the variance of the Gaussian noise added to each sample influences the resulting saliency map. Gaussian noise of zero mean and variances of 0.001, 0.01, and 0.1 are tested, and the results displayed in Fig. 8. Small Gaussian noise variance results in the saliency maps are too variable and lack features, seen in Fig. 8(a). Large Gaussian noise variance results in a more intense but indistinct saliency map, Fig. 8(c). Quantitatively, the interpretability metrics of each of the runs in terms of sparsity is 0.0954, 0.0654, and 0.0495, showing that as the noise increases it makes the resulting explanations flatter and more interpretable. The TV for each case is 0.0305, 0.0282, and 0.0285 respectively, meaning that once the added noise variance was increased to 0.01, the amount of noise in the explanation remained the same indicating that the interpretability stops improving. A balance of the Gaussian noise, 0.01 for this dataset, extracted the most features visually and balances the visual flatness and variance of the explanation for better interpretability, and should be used, see Fig. 8(b).

4.4.2. SmoothGrad parameter: number of noise samples

The number of noise samples in SmoothGrad, $N_{smthgrd}$, is a critical parameter. Too few samples result in a highly variable and sharp map, as in Fig. 9(a). Features are spread over a wide area, outlined in red. As more samples are added, the saliency map becomes more stable and features are visually distinct, as in Fig. 9(b), where the main features are outlined in magenta. At high numbers of samples however, features are smoothed out due to averaging and lose intensity and shape, as in Fig. 9 (c), where the features are outlined in orange. This adds significant computational time. Quantitatively, the interpretability metrics of each of the runs in terms of sparsity is 0.1027, 0.0646, and 0.0601, and in terms of TV, is 0.0812, 0.0305, and 0.0247. From a human interpretability perspective, it shows that as the number of noise samples

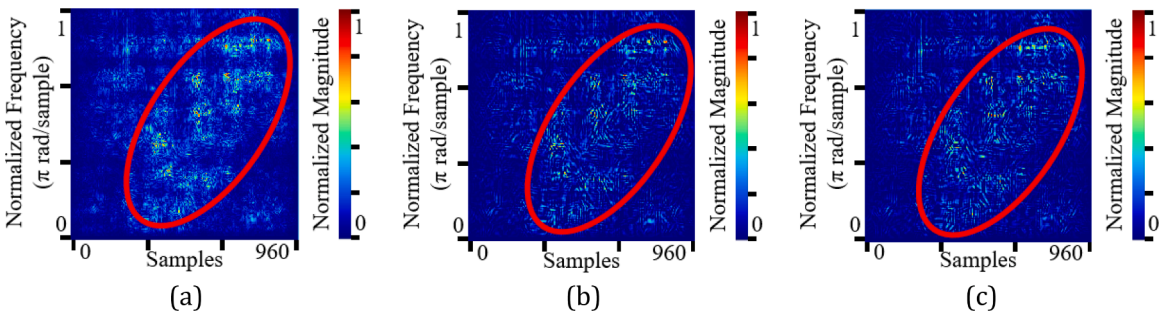


Fig. 6. The VG saliency maps generated when using the a) maximum absolute value, b) minimum value, and c) maximum value, of the gradient.

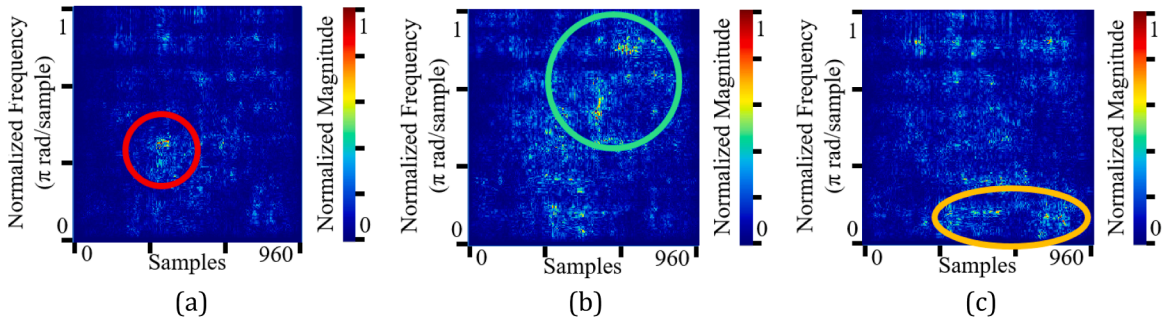


Fig. 7. A comparison between the VG saliency maps generated from three separate runs of the same DoS attack test case.

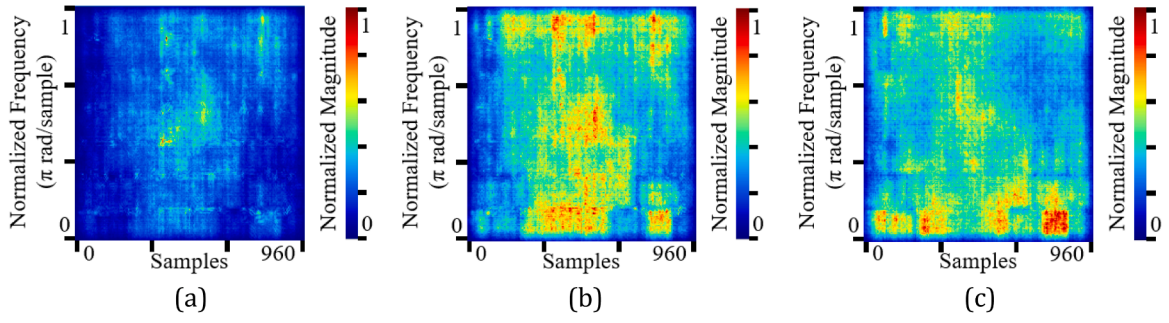


Fig. 8. The generated SmoothGrad saliency maps with Gaussian noise parameter settings of a) 0.001, b) 0.01, and c) 0.1.

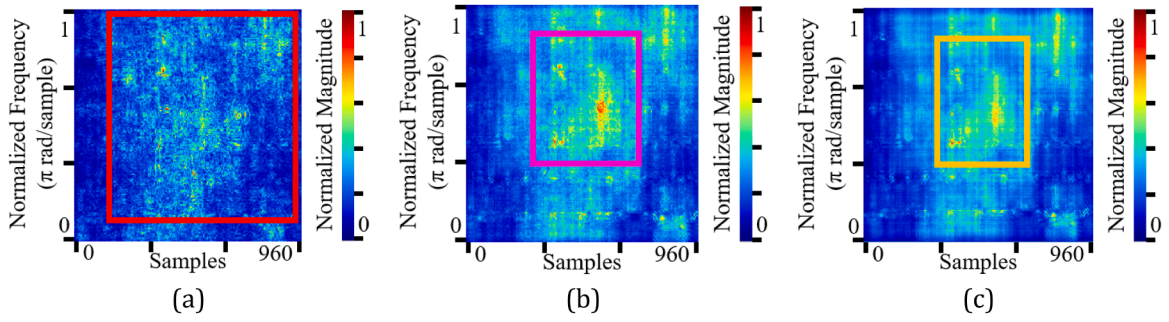


Fig. 9. The generated SmoothGrad saliency maps when a) 5, b) 50, and c) 500 noise samples are used for the explanation.

increases the density of the features in the image decrease, with the largest change between 5 and 50 samples. However, as the sparsity and TV continue decreasing, a loss of features is indicated as they are blurred. In terms of variability, as the number of samples is increased the variability decreases, this comes at the cost of increased flatness, which may cause key features to be lost. In terms of variability between runs, the MSE between two explanations of the same test case was measured to be 0.0104, 0.0016, and 0.0003 for the 5, 50, and 500 noise samples respectively. This shows that as the number of noise samples increases, the stability of the explanations improved. A balance between the stability, interpretability, and computational time for the used dataset was 50 noise samples.

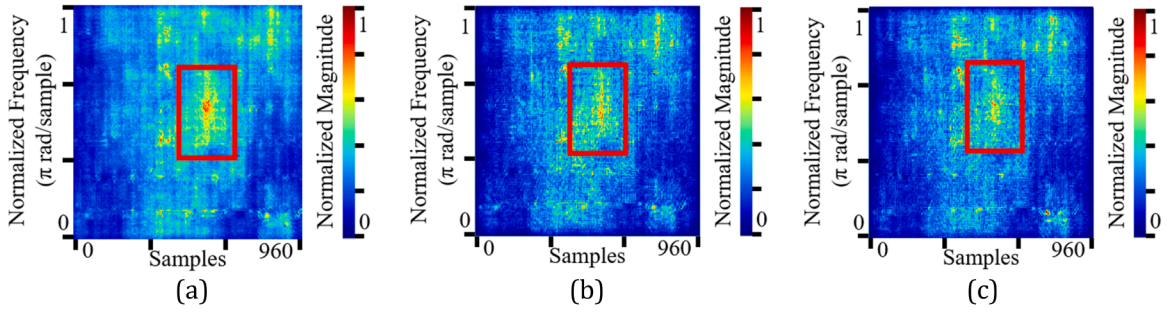


Fig. 10. The generated SmoothGrad saliency maps when the maximum of the a) absolute, b) negative, and c) positive, values of the gradient are used to generate the map.

4.4.3. SmoothGrad parameter: gradient representation

The resulting averaged gradients from SmoothGrad can be represented like gradients in VG: the maximum of the absolute values, the minimum values, and the maximum values. These are depicted for the DoS attack test case in Fig. 10. This method has the same behaviour as VG: the maximum of the absolute value provides the best visual features, as seen in Fig. 10(a), and the method can only identify regions of interest. The main feature in the explanations is outlined in red, and in Fig. 10(a) is the clearest and most defined. Quantitatively, the interpretability metrics of each of the runs in terms of sparsity is 0.0631, 0.0755, and 0.0722, and in terms of TV, is 0.0295, 0.0480, and 0.0475. From a human interpretability perspective, the absolute value of the gradient yielded a flatter explanation with the least variation and is best for interpretability, validating what is seen visually. It is notable that SmoothGrad cannot effectively identify features that positively or negatively affected classification.

4.4.4. SmoothGrad key issue: limited explanation meaningfulness

Like VG, SmoothGrad cannot distinguish what features helped or hurt classification, only what was important and as such its explanations are of limited meaningfulness.

4.5. Grad-CAM Analysis

4.5.1. Grad-CAM Parameter: feature layer selection

The layer of the CNN from which the gradients are taken from significantly influences the saliency map features. Early layers in the CNN contains higher-resolution features, refer to Fig. 11(a), where the features highlighted in red are clear and detailed. Moving through the CNN, pooling layers reduce the size of each layer but extract higher-level features, as in Fig. 11(b), where the feature highlighted in red has been reduced. As a result, the smallest layers are right before the softmax layer at the output of the CNN, but contain the final features used for detection. The feature in question here, outlined in red, has been reduced to a single corner without much detail, see Fig. 11(c). The features most relevant for explainability are those from the final layer, as such features are the highest-level combinations of preceding extracted features.

4.5.2. Grad-CAM Parameter: upsampling method

The output of Grad-CAM is a map that is smaller than the input image. Upsampling must be used to rescale the data to the resolution of the input image for comparison. Two options are bicubic upsampling, Fig. 12(a), and nearest upsampling, Fig. 12(b). Due to the low resolution of the raw explanation data, bicubic upsampling removes the sharp edges and smooths the resulting saliency map. This can best be seen around the main feature in the map, outlined in red in both Fig. 12(a) and (b). The smoothness of the two test cases can be measured using TV, where bicubic upsampling yields a TV value of 0.0032, and nearest upsampling a TV value of 0.0045. This indicates that the bicubic sampling yielded fewer overall variations, making it better for human interpretation. To avoid the sharp transitions introduced by the nearest sampling in this work, bicubic upsampling is used for Grad-CAM and the other XAI techniques that utilize upsampling.

4.5.3. Grad-CAM Key issue: low explanation resolution

Grad-CAM extracts data from the end layers of the CNN to most effectively identify what resulted in the final classification, but CNNs traditionally reduce the number of features through the network so these layers are small in dimension. This results in the final Grad-CAM map having a low resolution, an inherent weakness.

4.6. SHAP Analysis

4.6.1. SHAP Parameter: image segmentation method

As the number of features present in the image increase, SHAP suffers from significant computational costs. Therefore, image segmentation is used to reduce the image into a smaller set of features. This is achieved by dividing the input image into smaller areas using a grid and then averaging the pixels in each area to yield a smaller set of features. The grid's parameters are the number of rows and columns to break down the input image into. Values for these rows and columns are investigated in Fig. 13. As the number of rows and columns increase, so does the resolution of the resulting saliency map. With lower numbers of rows and columns features are lost, but too many rows and columns reduce the intensity of the found features. This can be measured in the sparsity value of the three explanations, which was 0.0838, 0.0359, and 0.0137 respectively. As the resolution of the saliency map increases, the features become weaker and more distributed, seen in how the sparsity value becomes flatter (decreases) as the resolution increases. The TV metric for each of the values is 0.0086, 0.0092, and 0.0071 respectively. As the TV values are low, the highest value should be taken as it indicates more changes in the explanation, and in this case, more detail, which is achieved with 14 rows and 32 columns. A balance between high resolution and interpretability must be selected, in this work 14 rows and 32 columns, from Fig. 13(b).

4.6.2. SHAP Key issue: assumption of feature independence

A key drawback of SHAP is that it assumes that each feature is independent [24]. However, in a SDS, features like physical currents and circuit breaker settings can be correlated, but from the data preprocessing this can result in such features existing in different sections of the spectrogram. With image segmentation, local regions of the input image are averaged, but the subsequent features are treated independently. This does not account for features correlated in the input image, but not adjacent to each other, and as such the explanations are missing features.

4.7. LIME Analysis

4.7.1. LIME Parameter: number of features

The number of features used to segment the input test case image

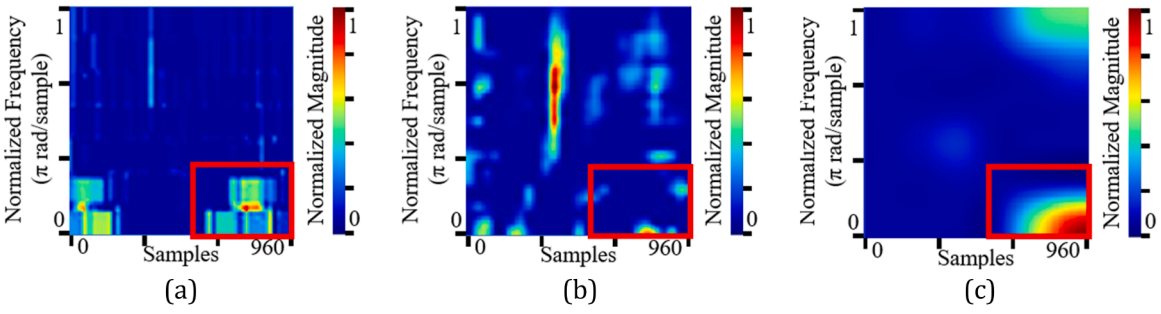


Fig. 11. The generated Grad-CAM saliency maps when extracting the feature layer data from a) the first convolution layer, b) the second ReLU layer, and c) the final layer before the softmax layer.

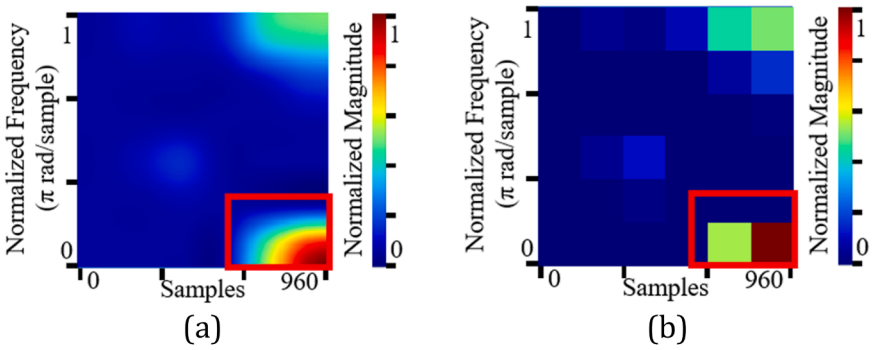


Fig. 12. The generated Grad-CAM saliency maps a) bicubic and b) nearest upsampling methods are used to upsample the explanations.

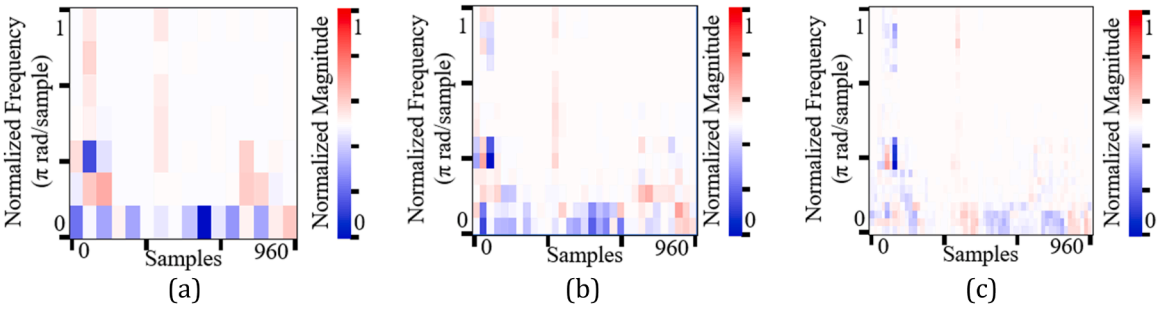


Fig. 13. The generated SHAP saliency maps when the input image is segmented into (a) 7 rows and 16 columns, (b) 14 rows and 32 columns, and (c) 28 rows and 64 columns.

greatly impacts the resulting saliency map. If too few features are chosen, such as 10 as in Fig. 14(a), then the model trained on samples is unable to capture the behaviour of the real model, resulting in a less meaningful explanation. Note how the features in Fig. 14(a) do not align with the features from (b) and (c). Too many features, such as 100, increases the computational time of the model and overfits to features that are not meaningful, refer to Fig. 14(c), where a new feature, outlined in red, was generated. Notably, the sparsity of each of the cases is 0.2494, 0.5336, and 0.5993, indicating that the features of interest are highly concentrated. In terms of TV, it was measured to be 0.0125, 0.0066, and 0.0072, respectively, showing that with 50 features the least variations visually are seen, and is the most interpretable by a human in this quality. A balance in the number of features, in this case, 50, balances between underfitting and overfitting the simpler model, while making the explanations the most interpretable in terms of variations, see Fig. 14 (b).

4.7.2. *LIME Parameter: segmentation method*

The methods used to divide the input image down into a smaller set of features in LIME are segmentation using Superpixels and grid segmentation. Fig. 15(a) shows the effect of Superpixels, which vary their shape based on the available features, capturing non-uniformly shaped features, while also keeping a high resolution with a low number of features. The main feature is outlined in red. On the other hand, the grid method in Fig. 15(b), does not adapt to the shape of the features and the resolution is determined by the number of features. This leads to spurious features being identified on top of the main feature already highlighted in red, with notable spurious features outlined in magenta in Fig. 15(b). With a low number of features, the grid segmentation method does not represent complex features well. To best represent features, the Superpixels method should be used.

4.7.3. *LIME Parameter: number of synthetic images*

This number directly influences the training of the simpler model. With too few synthetic images the variability of features between runs of the model is high. For the test dataset, the significant features begin stabilizing at 1024 images and stop changing at 2048 images. At 4096 images, the significant features no longer change, but there is a significant increase in the required computation time. The resulting feature importance maps for these numbers of synthetic images are shown in Fig. 16. The MSE differences between the first case and second case is 0.0041, while from the second case to the third, the MSE is 0.0015. For a

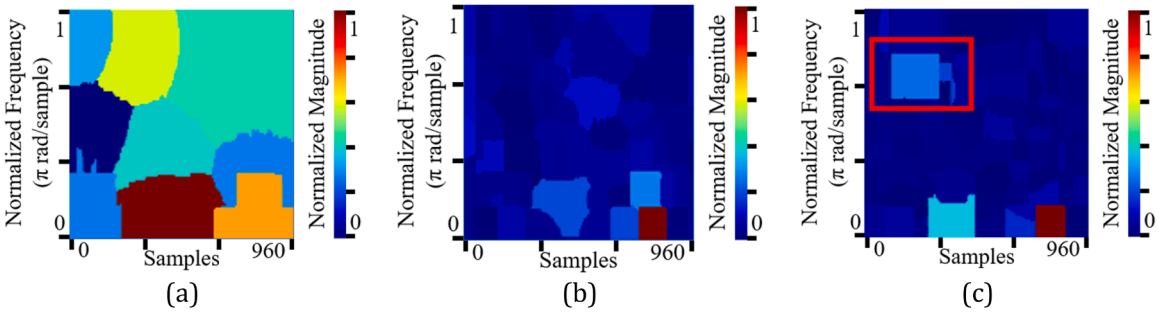


Fig. 14. The generated LIME feature importance maps when the following numbers of features are used: a) 10, b) 50, and c) 100.

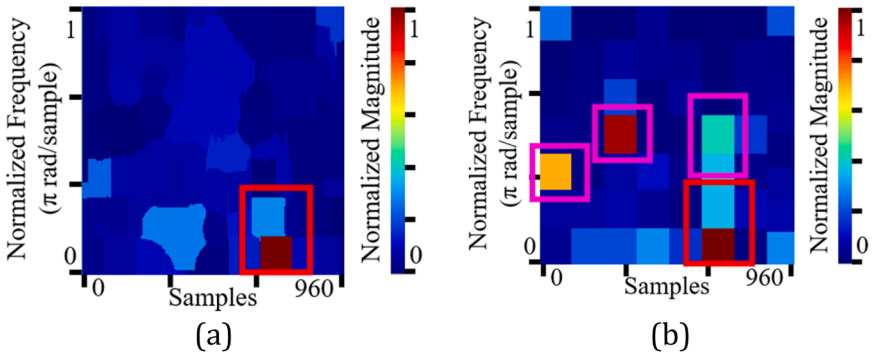


Fig. 15. The generated LIME feature importance maps when, a) Superpixels segmentation, and b) grid segmentation is used.

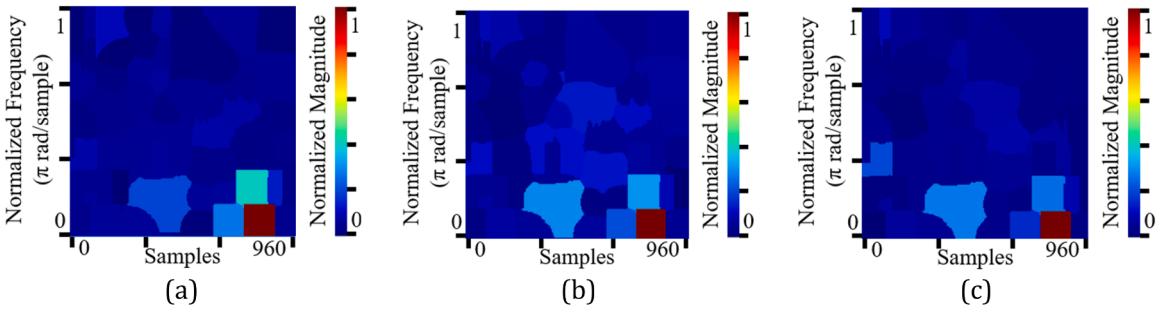


Fig. 16. The generated LIME feature importance maps when using the following number of synthetic images: a) 1024, b) 2048, and c) 4096.

significant increase in computation time from the second case to the third, a change much smaller is seen than from the first to the second, indicating the diminishing return. No additional features of significance were extracted, so the additional samples and computational time was not necessary. The number of synthetic images should be selected to balance the computational time and will vary based on the dataset and models used. For the given dataset 1024 was taken as the balance.

4.7.4. LIME Parameter: model type

In this work, Decision Tree (DT) and a Linear Regression model are compared in Fig. 17. It is seen that visually the DT model is better able to extract the most important feature, seen in Fig. 17(a), while the linear model assigns a high performance to most of the image as nearly the entire map in Fig. 17(b) has a normalized magnitude above 0.5. The resulting explanation sparsity when using each model is 0.5206 and 0.0234 respectively, indicating that the Linear Regression model has much less concentrated features and can be seen to be more visually confusing in Fig. 17b). The TV of the DT-based explanation is 0.0072, and for the Linear Regression-based explanation is 0.0112, indicating that the DT-based explanation is more interpretable as it has significantly less visual variance. For more interpretable explanations, the DT model should be used.

4.7.5. LIME Key issue: model complexity

It is notable that LIME trains a simple model on the specific inquired test case to explain the more complex ML model. This is limited in that the simpler model attempts to fit itself to the randomized samples and classifications. Inherently, it does not explain what precisely led to the decision made by the more complex ML model. This leads to questionable trustworthiness of the generated feature importance map, as a simple ML model like a DT cannot capture the complex behavior of an advanced ML model like a CNN.

4.8. OSM Analysis

4.8.1. OSM Parameter: stride and mask size

The stride parameter refers to the step size used to iteratively move the occluding mask across the input image. Each OSM sample generated slides the occluding mask horizontally and vertically in increments of the stride. The mask size parameter refers to the dimensions of the mask.

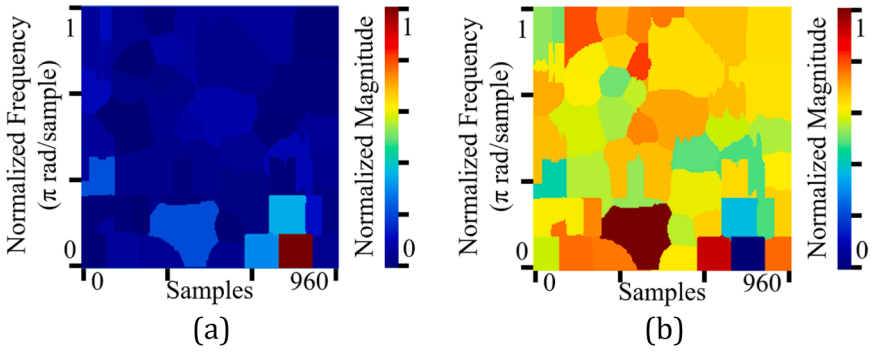


Fig. 17. The generated LIME feature importance maps when using: a) a DT model, and b) a Linear Regression model.

In this work the mask is limited to a square shape, where the mask size refers to the height and width dimensions of the mask in pixels as in Fig. 18.

In this work, the input image size is 227 by 227. The mask size and stride were varied across the range of 20 and 220 to provide an analysis across the range of possible sizes. A figure showcasing the impact of the choice of mask size and stride is given in Fig. 17. It is evident that the smaller stride and mask sizes provide more useful features. Once the stride size exceeds 60 pixels there are too few masks to generate results, and the resulting saliency maps have low resolution and less meaningful results. A larger mask size also occludes too much of the image at once, resulting in less useful feature maps.

It is notable that small stride and mask sizes generate a high number of images and result in significant computational time. Additionally, small masks do not occlude enough of the input image to extract useful features. A balance between the stride and mask size was found experimentally to be a mask size of 20 and a stride size of 10. This allows the mask to be slid across the image, whereas the smaller stride size allows each iteration of the occluding mask to overlap the last, thereby allowing the features on the edge to be included in the center of the next mask, reducing the effects of the sharp boundaries. Stride size is related to the computational complexity of OSM, as the number of samples generated that need classification is directly related to the stride size. The number of samples evaluated to form the saliency map when mask clipping is not enabled is given by (7) for a square input image, where N_{image} is the width of the square image, and N_{stride} is the used stride size.

$$N_{OSM\ Samples} = \left\lfloor N_{image}/N_{stride} \right\rfloor^2 \tag{7}$$

4.8.2. OSM Parameter: mask clipping

In OSM the mask clipping parameter dictates how masks are applied at the edges of the input image. With mask clipping, masks cannot overlap with the region outside the image; masks that exceed the image dimensions are ignored and not used for analysis. As is seen in Fig. 19, mask clipping reduces the features on the edges of the image as they decrease in shape and lose definition, highlighted in red. Other features, highlighted in magenta, appear out of nowhere in Fig. 19(b). Due to the loss of features and generation of spurious features mask clipping should not be enabled to retain the features on the edge of the image.

4.8.3. OSM Parameter: mask replacement value

The mask replacement value parameter in OSM refers to the value of the mask used to occlude the features in the input data. The value was varied across the full image range of 0 to 1, with examples provided in Fig. 20. The input image's values were already scaled to be between 0 and 1. At low mask replacement values, low magnitude features, as that in outlined in magenta in Fig. 20(a), contain less detail. As the mask replacement value is increased to about the middle of the range, these features become more defined, see the highlighted magenta feature in Fig. 20(b).

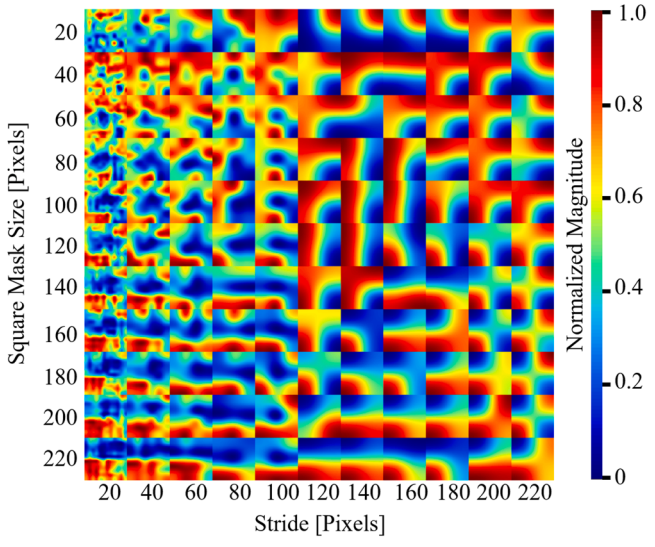


Fig. 18. A visualization of the impact of the stride size and mask size on the resulting OSM maps.

At high mask replacement values the features visually lose intensity and some noise is added, refer to the highlighted magenta feature in Fig. 20(c). The sparsity of each of the three explanations is 0.0516, 0.0513, and 0.510, while the TV values are 0.0137, 0.138, and 0.0138. The choice of mask value in this case does not significantly alter the interpretability of the explanation, so it should be selected to instead improve the visual quality of the features.

To balance the details of the features and their intensity a mid-range value should be selected. In this work the mask replacement value was selected to be that of the average pixel value from the original test image.

4.8.4. OSM Key issue: instability with respect to parameters

It is notable that the OSM maps generated are highly dependent on the parameters used, specifically stride and mask size as shown in Fig. 18. This indicates that not enough occluded mask samples are used to generate the saliency map.

The resulting explanations are therefore dependent on the parameters of stride and mask size, limiting the trustworthiness of the saliency maps generated by this methodology as completely different explanations can be created by slight changes in parameter selection.

4.9. RISE Analysis

4.9.1. RISE Parameter: generated mask size

The size of the generated mask of ones and zeros initially before being scaled up affects the resulting size of occluding patches. A small relative generated mask size, such as 1% in Fig. 21(a), results in large occlusion patches that fail to extract many features. A large relative generated mask size, like 10% in Fig. 21(c), results in higher-resolution features, but a noisy mask generates more random explanations due to the increased number of combinations of occluded features.

A generated mask size of 5%, shown in Fig. 21(b), was used as it balances the resolution of the resulting saliency map and the randomness of the explanations. This is confirmed with the sparsity values of the resulting explanations, of 0.1269, 0.0306, and 0.0634 respectively, and of the TV metric, 0.0044, 0.0096, and 0.0163.

The sparsity was the lowest value for the 5% mask value, indicating that the features were less concentrated than the other cases, and the TV shows that as the mask size increases the amount of variation in the explanation increases. This concurs the chosen value in this work.

4.9.2. RISE Parameter: percentage of input image to occlude

The percentage of the input image to randomly occlude in every sample is a parameter that can be varied from 0% to 100%. Parameter settings of 25%, 50%, and 75%, are depicted in Fig. 22. Visually, 50% yielded the least noisy result. When only a small part of the input image is occluded per sample additional samples are required to adequately cover the input image space. When the number of samples is kept the same, but the amount of occlusion increases, the input space is not equally covered by the masks. This results in more variable results and explanations with significant noise, as outlined in red in Fig. 22(a), where the entire explanation is covered with features. With too much occlusion too little of the original image is used for classification, so classification is mostly performed using the occluded components, resulting in a saliency map that is not meaningful, leading to Fig. 22(c), where several key features are scatted across the image, with two outlined in orange. Additionally, the sparsity of the explanations for each parameter value are 0.0461, 0.0661, and 0.0498, and the TV metric

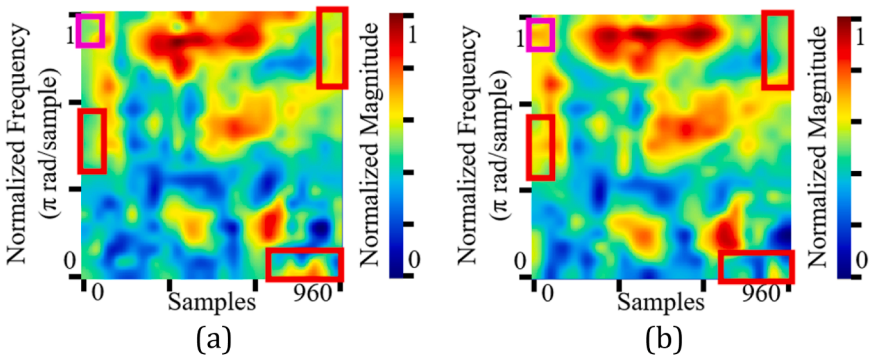


Fig. 19. The generated OSM maps when mask clipping is a) disabled, and b) enabled.

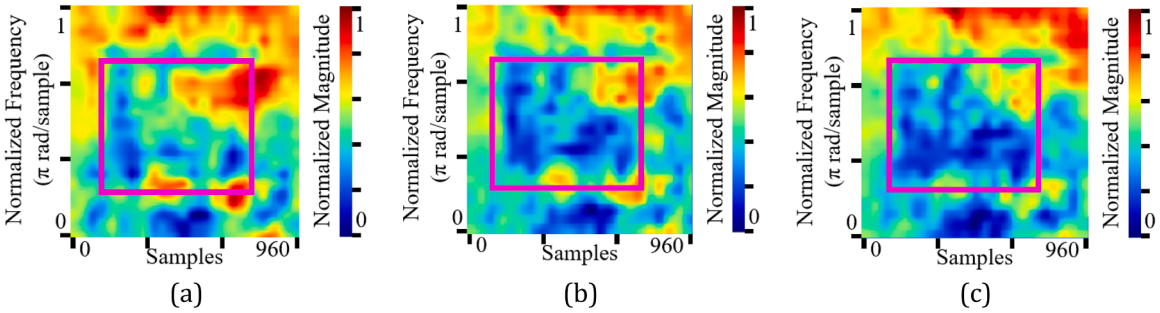


Fig. 20. The generated OSM maps when the mask replacement values is a) 0, b) 0.5, and c) 1.

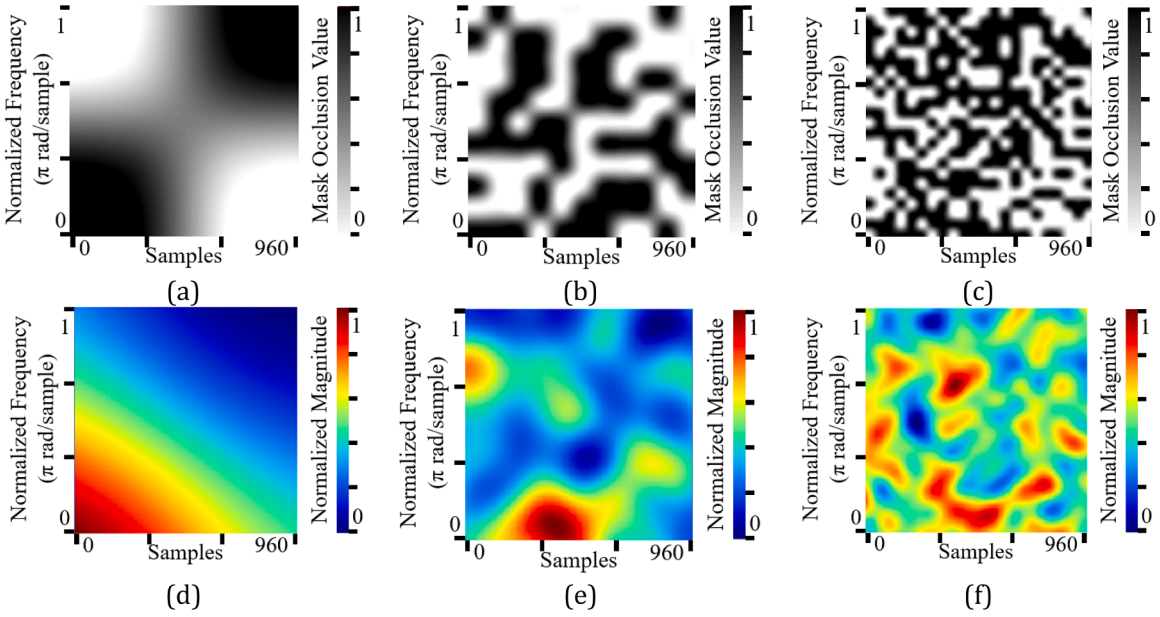


Fig. 21. The generated RISE masks with generated mask sizes of a), 1%, b) 5%, and c) 10%, of the input image, and the corresponding generated RISE maps, d), e), and f), when the respective generated mask sizes are used.

0.0102, 0.0090, and 0.0498 respectively. This shows that the 50% occlusion value produces the most concentrated features as it has the highest sparsity value, while it has the lowest amount of TV, meaning it will be the most interpretable of the three options as it is the smoothest. As such, a balance of 50% was experimentally determined to yield the most meaningful and interpretable results, as in this test case a single clear key feature, outlined in magenta in Fig. 22(b), was seen.

4.9.3. *RISE Parameter: number of samples*

The number of utilized RISE samples influences the stability of the resulting saliency maps. By measuring the evenness of the coverage of the input image by the generated masks an insight to the stability of the saliency maps can be determined. It is desired that when averaged, the masks result in equal occlusion across the image. This avoids the explanation focusing on specific locations of the image. To achieve a numerical measure of this equality, the generated RISE masks are averaged and the MSE of the resulting averaged mask is taken, giving the

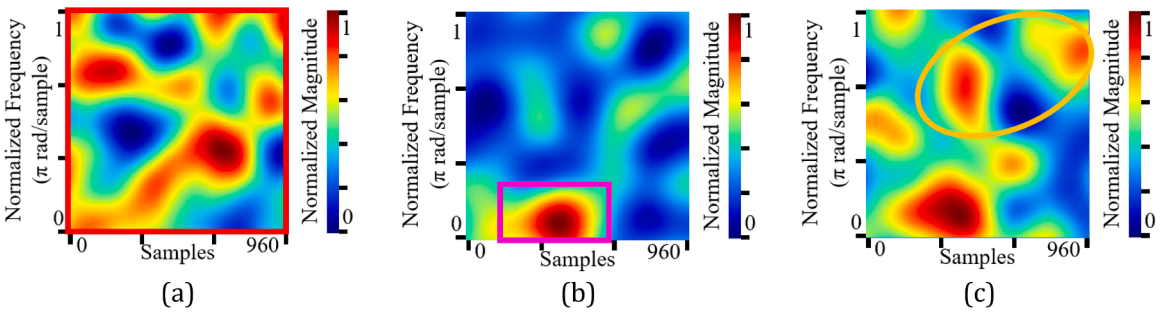


Fig. 22. The generated RISE maps when the percent of the input image that is occluded parameter is set to a) 25% b) 50%, and c) 75%.

maskMSE. The *maskMSE* is calculated as in (8), where there are *PX* total pixels in the image of the averaged mask, and px_{avg} is the average value of a pixel in that image. The MSE of the averaged mask compared to the number of RISE samples is plotted in Fig. 23, which shows that the MSE decreases linearly on a log-log scale as the number of samples increase. As the number of samples increase the masks cover the input space more evenly.

$$maskMSE = \sum_{pixel\ px\ in\ PX} (px - px_{avg})^2 \tag{8}$$

To further inform what number of samples results in adequate coverage of the input image, the maximum and minimum values of the averaged mask is shown in Fig. 24. For the given test system diminishing returns are seen after 1000 samples, and after 10 000 there is negligible change. To balance the number of samples and the resulting computational complexity between the stability of the results, with the given system and parameters the number of RISE samples was selected to be 1000 for better time performance.

4.9.4. RISE Key issue: computational complexity

The main limitation of RISE is that a significant number of samples are required to calculate a stable saliency map. For this dataset, a minimum of 1000 samples are required. Compared to the other XAI methods which use sampling, this is a significant increase. However this is also a strength of RISE, as that because of the large number of samples the explanations are less variable, and are repeatable.

5. Comparison of XAI techniques

An overview of the analyzed XAI Techniques is provided in Table 6. The metrics chosen are related to the applicability of the desired application of XAI to a SDS CPS, where trustworthiness of the methodology is key. The comparison is based on the experimental results from testing each XAI technique and its parameters. Specifically, for the "Stable (Repeatable) Explanations" metric, it considers how much the explanations change for a given XAI when the parameters are modified,

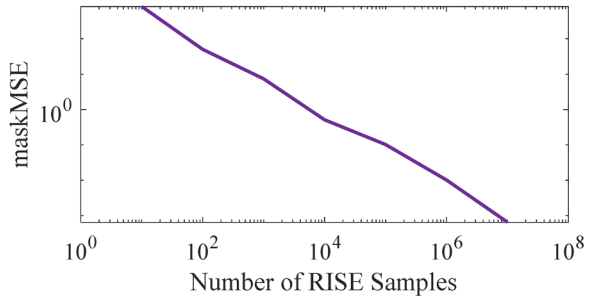


Fig. 23. A semi-log plot of the *maskMSE* metric to the number of RISE samples used, showing that the mask MSE decreases semi-logarithmically with an increase in the number of RISE samples.

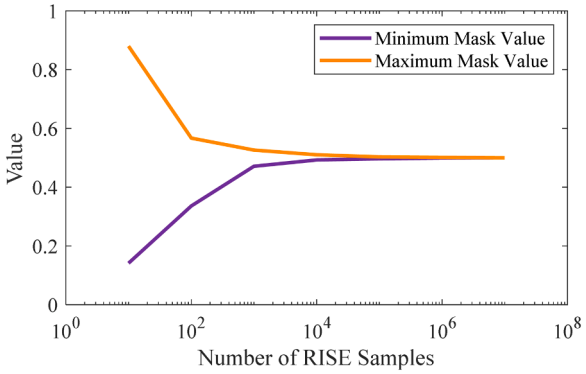


Fig. 24. A semi-log plot of the maximum and minimum values in the averaged RISE mask compared to the number of RISE samples generated, showing that the values converge as the number of RISE samples increase.

Table 6
Qualitative comparison of the analyzed XAI methodologies.

XAI Technique	Model-Agnostic	Handles Dependent Features	High Resolution Explanations	Stable (Repeatable) Explanations
VG	×	✓	✓	×
SmoothGrad	×	✓	✓	×
Grad-CAM	×	✓	×	✓
SHAP	✓	×	✓	✓
LIME	✓	×	✓	✓
OSM	✓	✓	✓	×
RISE	✓	✓	✓	✓

as for some methods a slight change results in completely different explanations. From the given comparison metrics, it is seen that RISE outperforms to the other methodologies in the given experimental setup as it satisfies all the metrics in Table 6 and results in the most trustworthy methodology. Since RISE utilizes random sampling with many samples it is much more stable between runs and as is seen from the analysis of the convergence of MSE of its masks, lowers statistical uncertainty through the evaluation of additional data points.

A summary of the key drawbacks and benefits of each method is provided in Table 7. As computational complexity is also a critical factor when attempting to apply XAI in a time-sensitive scenario, Table 8 identifies that largest source of complexity in each XAI method. The computational complexity of each method is further investigated in Section 5.3. Quantitative analysis results are provided in the following sections. To accurately compare the XAI methods both the quantitative and qualitative results must be considered.

5.1. Limitations across XAI methodologies

This section reviews the key identified limitations of each XAI

Table 7
Highlights of each XAI methodology.

Methodology	Key Drawback	Key Benefit
VG	Results are highly variable	Very fast
SmoothGrad	Only can identify points of interest, and not if regions helped or harmed classification	Simple implementation
Grad-CAM	Low resolution explanations	Is integrated directly with the CNN for fast performance
SHAP	Assumes feature independence	Robust to the selection of parameters
LIME	Limited capture of the model's behaviour	Relatively robust to the selection of parameters
OSM	Highly variable with respect to its input parameters	A fast model-agnostic method
RISE	Requires many samples; can be computationally demanding	Stable and repeatable explanations

Table 8
Sources of the computational complexity of each XAI methodology.

Methodology	Main Source of Complexity
VG	A single forward pass of the CNN
SmoothGrad	Forward passes of the CNN required for each sample
Grad-CAM	One forward pass of the CNN, followed by backpropagation
SHAP	Requires forward passes of the model based on the number of features
LIME	Requires forward passes of the model for the number of representative samples, followed by the training of a simple ML model
OSM	Forward model passes are required for each sample
RISE	Forward model passes are required for each sample, is necessary to generate large numbers of randomized masks

methodology and compares them across the investigated methods. Several methods suffer from variability between runs or with its parameters, like VG and OSM. In practice this can be mitigated by picking techniques that are less variable, including RISE and Grad-CAM, to ensure that the explanations are reproducible and meaningful. Gradient-based methods like VG and Smoothgrad return explanations of limited use as they cannot highlight regions that positively or negatively affected classification, only what was of interest. For more useful explanations the other methods should be used. Low-resolution explanations are uniquely encountered with Grad-CAM; the other methods do not suffer from this and as such another XAI methodology can just be chosen. Inaccurate explanations can be generated by some methods due to their limitations in capabilities, like SHAP with its inability to consider feature independence, and LIME with its modelling of only local behaviours of a model around a test case using a simple ML model. This issue of inaccurate explanations is not suffered by the other XAI methodologies, which can be used as a substitute. Computational complexity affects all methods, but more heavily affects some methodologies, like RISE and SHAP. Notably, the overall quality of explanations appears to be inversely related to computational complexity, as methodologies that are fast lack meaningful explanations, like VG, while methods like RISE have more meaningful explanations but are slower. This is a key trade-off that should be considered since it cannot be completely avoided. No method solves each limitation, but some methods outperform others for use cases. By using the above guidance more appropriate XAI methodologies for a use case can be selected.

5.2. Quantitative comparison of the XAI methodologies

In this section the XAI methodologies investigated are compared quantitatively. The methodologies are analyzed using interpretability, stability, faithfulness, and fairness as outlined in Section 4.1.

5.2.1. Quantitative comparison: interpretability and stability

The human-interpretability, measured by sparsity and TV, and the stability of each method for each XAI methodology is given in Table 9. It is seen that the method with the lowest sparsity is SHAP, which returns the flattest images with the most distributed features. The method with the highest sparsity, and therefore the most concentrated explanations, is LIME, which is due to how LIME assigns Superpixels values and leads to concentrated regions of uniform magnitudes in its explanations. The lowest TV and therefore the least noisy explanations, is achieved by Grad-CAM. This is explained by the bicubic sampling of a low-resolution image, leading to large smooth regions. RISE, OSM, SHAP, and LIME are all close in range, indicating that with the parameters selected they all yield similarly noisy explanations. VG and SmoothGrad perform the worst in this regard, as they depict gradient data, which varies quickly and widely. As for stability, SmoothGrad performed the best, improving upon the second-best VG, indicating that the averaging of the gradients improves the resilience to noise. The worst performing methodology for stability is LIME, indicating that with slight noise the simpler model within LIME represents the behaviour of the CNN worse. Both occlusion-based methodologies, RISE and OSM, yield similar stability results with only a 0.0011 difference. The results of this analysis must be considered with the other investigated behaviours of each method in this work, as the results in this section are averaged across the entire testing dataset.

5.2.2. Quantitative comparison: fairness

This section analyzes the results from testing the metrics across different attack types, instead of using only a single DoS attack case, which was used previously for analysis, to measure fairness as outlined in Section 4.2.4. The averaged sparsity, TV, and stability metrics are given for each class, as well as the mean of the per-class metrics and the Sample Variance of the per-class metrics across the dataset, in Table 10.

The table shows that each of the metrics do not vary significantly across the classes in each dataset as per the low Sample Variances, signifying that the explanations represent each class fairly. The most variance is seen in the AUC metric, which measures the drop in the model confidence as the highest-identified pixels to classification are removed, and measures the response of the model to the synthetic deleted images of which there is some noise present. However, the values are all within an order of magnitude. Since the metrics across the classes do not change significantly it is seen that the explanations fairly treat each class. Table 11 compares the average quantitative metric for each class against the averaged value across the entire dataset, providing insight into whether single classes or metrics differ the most from the averages, indicating a worse measure of fairness. To aid the analysis, the differences from each of the corresponding metric's means are summed class-wise and metric-wise and then normalized. As indicated by the normalized average difference from the mean metric per-class, the Disturbance class varies the least highlighting that this class was treated fairly. The worst performing class is the FDI attack class, but it lies within 10% of the next class in this metric and is not significantly larger than the other classes. By class, the XAI explanations are fair across all metrics aggregated and do not significantly misrepresent any single class.

As for the normalized average difference from mean metric per-

Table 9
Interpretability and stability metrics of each XAI method across the entire testing dataset.

XAI Technique	Sparsity	TV	Stability
VG	0.2666	0.0389	0.0055
SmoothGrad	0.0692	0.0293	0.0026
Grad-CAM	0.2167	0.0065	0.0340
SHAP	0.0255	0.0097	0.0073
LIME	0.5357	0.0106	0.0582
OSM	0.2585	0.0089	0.0393
RISE	0.0852	0.0087	0.0378

Table 10
An analysis of the fairness of RISE in terms each class in the testing dataset.

Class	Sparsity	TV	Stability	AUC
Normal	0.2549	0.0576	0.0048	0.2360
Disturbance	0.2358	0.0532	0.0053	0.1445
Attack, FDI	0.2386	0.0580	0.0052	0.2493
Attack, Replay	0.2403	0.0548	0.0056	0.1153
Attack, MS	0.2590	0.0525	0.0048	0.1106
Attack, DoS	0.2585	0.0481	0.0040	0.0993
Attack, Insider	0.2431	0.0605	0.0050	0.2183
Geometric Mean	**0.2472**	**0.0550**	**0.0050**	**0.1676**
Sample Variance	**9.909e-5**	**1.720e-5**	**2.620e-7**	**4.184e-3**

Table 11
An analysis of the fairness of RISE in terms of the difference between the aggregated metric for each class and the geometric mean across all classes.

Class	Difference from Metric Mean				Normalized Average Difference from Mean Metric, Per-Class
	Sparsity	TV	Stability	AUC	
Normal	0.0077	0.0026	−0.0002	0.0684	0.0197
Disturbance	−0.0114	−0.0018	0.0003	−0.0231	0.0092
Attack, FDI	−0.0086	0.0030	0.0002	0.0817	0.0234
Attack, Replay	−0.0069	−0.0002	0.0006	−0.0523	0.0150
Attack, MS	0.0118	−0.0025	−0.0002	−0.0570	0.0179
Attack, DoS	0.0113	−0.0069	−0.0010	−0.0683	0.0219
Attack, Insider	−0.0041	0.0055	0.0000	0.0507	0.0151
Normalized Average Difference from Mean Metric, Per-Metric	0.0088	0.0032	0.0004	0.0574	

metric, there is more variance. The value for stability is eight times less than the next largest value, showing that stability across each class is more similar than the other metrics, indicating that this is the fairest metric. The worst is AUC at 0.0574. However, in the context of this metric, which varies from 0 to 0.25, this represents a shift of up to 23% of the mean of the metric. However, this metric is an integration from the AUC plot from which variations are expected between cases. However, for each of the metrics across all classes, there are variations, but no class is distinctly mistreated or favoured, highlighting fairness. To verify the fairness of each XAI methodology the Sample Variance of the sparsity, TV, stability, and AUC metrics are compared in Table 12. This table shows that all the XAI techniques have low Sample Variances for the metrics, indicating that they are fair. However, some methods outperform others on specific metrics. RISE and SmoothGrad are the fairest in terms of sparsity, while SmoothGrad has the fairest TV, followed by Grad-CAM. For stability SmoothGrad, RISE, and VG were the fairest by orders of magnitude compared to the other techniques. For AUC many of the methods performed similarly in terms of fairness, with

Table 12
An analysis of the fairness of each XAI technique using the Sample Variance for metrics.

XAI Technique	Sample Variance			
	Sparsity	TV	Stability	AUC
VG	1.902e-4	2.691e-5	5.993e-7	3.857e-3
SmoothGrad	2.739e-5	5.981e-7	1.231e-7	4.347e-3
Grad-CAM	2.064e-3	1.213e-6	2.247e-4	1.987e-3
SHAP	3.135e-4	1.258e-5	7.335e-5	6.497e-3
LIME	3.391e-3	4.337e-6	2.451e-4	1.609e-3
OSM	2.252e-2	9.520e-6	3.874e-4	1.960e-3
RISE	9.909e-5	1.720e-5	2.620e-7	4.184e-3

LIME performing the best, followed by OSM and Grad-CAM. Since no technique significantly outperforms another in terms of all metrics of fairness, fairness should just be used as a verification of the performance of a chosen XAI methodology.

5.2.3. Quantitative comparison: faithfulness

The deletion plots for a DoS case used for the analysis of the XAI parameters in Section 4, is shown in Fig. 25 for each of the given XAI methods. The steep drop in model confidence highlights that the explanations are faithful representations. While models like VG and SmoothGrad, in Fig. 25(a) and (b), drop off the fastest, this is as they represent specific gradient data, and is not very interpretable for humans. While SHAP and RISE have much slower drop offs they still decrease as more of the salient pixels are removed, highlighting that they still are faithful explanations. These methods trade the steepness of the model confidence with respect to the percent of replaced pixels for more interpretable explanations.

The averaged AUC value for each methodology, averaged over the entire dataset, is given in Table 13, revealing that all methods return an average AUC of the deletion plot of a similar range, but some methods outperform others. As a lower AUC score is better, SHAP performed the worst indicating that due to its inability to account for dependent features the faithfulness of its explanations suffers. RISE performs the best, followed closely by OSM.

These model-agnostic methods, through occlusion, can better identify the important features. RISE performs better than OSM, which is logical as RISE, through many random samples, can better determine the key areas of the input spectrogram.

5.3. Statistical confidence measures of quantitative evaluation metrics

To provide more confidence on the reported quantitative evaluation metrics across each XAI method, the 95% Confidence Interval (CI) is used as these measurements can be modelled as binomial experiments and the standard Z-test holds [35].

For a given evaluation metric whose value is represented by x_m, the 95% CI, $CI_{95\%}$, of the value is determined using the Z-score as given by (8). For this $Z = 1.96$. The total number of cases in the dataset is represented by N_{cases}.

$$CI_{95\%} = \frac{2N_{cases}x_m + Z^2 \pm Z\sqrt{Z^2 + 4N_{cases}x_m - 4N_{cases}x_m^2}}{2\left(N_{cases} + Z^2\right)} \tag{9}$$

The results of the confidence interval analysis are provided in Table 14. It is seen that each metric lies within a 95% CI. The CIs for the metric do not signify significant changes or meaningful variations. Across the CIs each of the metrics vary, but not enough to significantly alter the quality of the explanations. As such, the CIs corroborate the conclusions drawn in the analysis.

5.4. Experimental runtime computational performance comparison

Each of the XAI methods were tested terms of the time taken to calculate a single explanation for a single test case, that is, the runtime of each XAI method. Tests were conducted on a computer consisting of an Intel i7-14700K processor, an NVidia A4000 Graphics Processing Unit (GPU), 16 GB of memory, and MATLAB 2024b. The results are provided in Table 15.

The computational time performance numbers are in line with the computational time guidance provided. VG is the fastest method by far, at the cost of its meaningfulness as previously described. While RISE is strong, it suffers from a need for many samples, leading it to have the most significant computational overhead. SHAP and LIME are also relatively computationally heavy, due to the number of samples required in SHAP, and the need to generate samples and train a new model with LIME. While SmoothGrad and OSM have relatively low

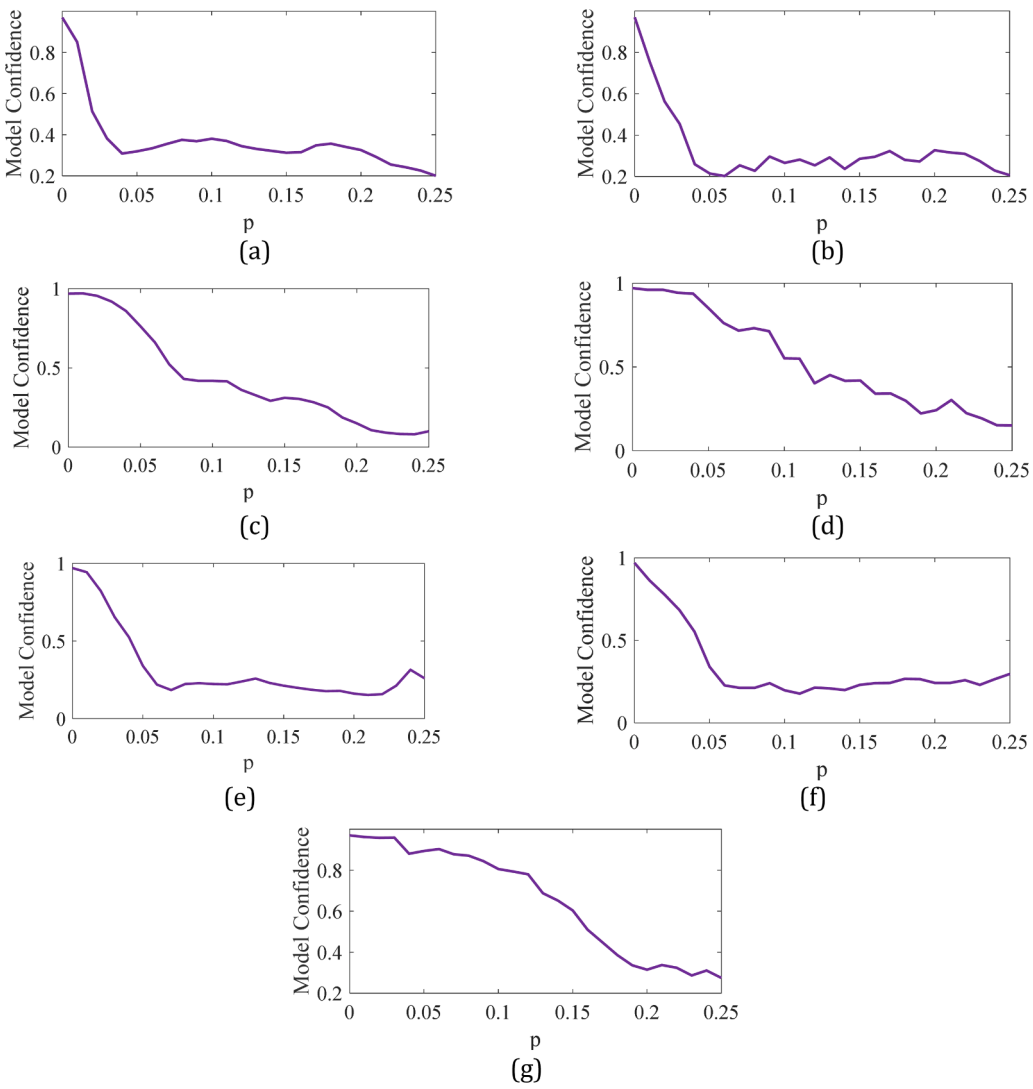


Fig. 25. The deletion plots for a single DoS attack case for each of the XAI methodologies: a) VG, b) SmoothGrad, c) Grad-CAM, d) SHAP, e) LIME, f) OSM, and g) RISE.

Table 13
XAI methodology faithfulness as measured using the average AUC of the deletion plots.

XAI Technique	Average AUC
VG	0.1705
SmoothGrad	0.1681
Grad-CAM	0.1627
SHAP	0.1863
LIME	0.1500
OSM	0.1420
RISE	0.1409

computational times, they are hindered by the quality of their explanations as discussed previously.

For real-time deployment, the suitable method depends on the required frequency of explanations. For a rapid, constantly-monitoring SDS the current XAI methodologies are more limited, with VG being the only suitable method as it can generate explanations in the order of tens of milliseconds. If explanations are necessary on the order of seconds, then more meaningful but computationally efficient XAI methods can be used like Grad-CAM and OSM. Once the explanations are necessary in the order of tens of seconds then all XAI methodologies apply. For example, if continuous explanations are necessary, but classifications are made only twice per minute, then the more meaningful RISE methodology can be utilized. The more computationally demanding XAI methods like RISE can be used for the faster applications however if sampling is used and explanations of all classified scenarios are not necessary.

For significant events of interest being analyzed after the fact, the fine time-resolution explanations can be generated at the desired time of the operator, and the runtime performance of the methods is less critical. If explanations are not required in real-time, but for future use, then the runtime performance of the methods is less critical and more meaningful methodologies, like RISE, should be used. Future work can focus on reducing the computational complexity of the XAI methods to improving the runtime for the methods, making faster explanations for SDS application available. It is also important to note that to not impede the rapid classification of an IDS the XAI explanations should be decoupled from the classification process, this can be achieved by performing the classifications in parallel.

5.5. *Generalizability*

The analysis thus far has focused on a single dataset. As analyzed in Section 5.2.2, RISE can explain test cases from each class fairly. To verify that the results are generalizable, a second dataset was generated from

Table 14
The 95% confidence intervals of the quantitative evaluation metrics for each XAI methodology.

Metric		VG	SmoothGrad	Grad-CAM	SHAP	LIME	OSM	RISE
Sparsity	Value	0.2666	0.0692	0.2167	0.0255	0.5357	0.2585	0.0852
	CI+	0.3121	0.0985	0.2598	0.0461	0.5841	0.3037	0.1167
	CI-	0.2255	0.0482	0.1790	0.0140	0.4866	0.2179	0.0616
TV	Value	0.0389	0.0293	0.0065	0.0097	0.0106	0.0089	0.0087
	CI+	0.0627	0.0509	0.0204	0.0251	0.0263	0.0239	0.0236
	CI-	0.0239	0.0167	0.0021	0.0037	0.0042	0.0033	0.0032
Stability	Value	0.0055	0.0026	0.0340	0.0073	0.0582	0.0393	0.0378
	CI+	0.0189	0.0142	0.0567	0.0216	0.0857	0.0632	0.0614
	CI-	0.0016	0.0005	0.0202	0.0024	0.0391	0.0242	0.0231
Average AUC	Value	0.1705	0.1681	0.1627	0.1863	0.1500	0.1420	0.1409
	CI+	0.2106	0.2080	0.2022	0.2275	0.1884	0.1797	0.1785
	CI-	0.1367	0.1346	0.1297	0.1511	0.1183	0.1111	0.1101

Table 15
A comparison of the required computational time, the runtime, of each XAI method.

XAI Technique	Computational Time [s]
VG	0.0540
SmoothGrad	2.5584
Grad-CAM	1.0110
SHAP	7.9704
LIME	6.9312
OSM	1.2110
RISE	8.8107

the test system in [28]. The exact test system is used with the same test scenarios and parameters. New data was collected and preprocessed into a new dataset using the exact process in [28]. This new dataset, called the validation dataset in this work, contains more test cases than the first dataset. A breakdown of the validation dataset is provided in Table 16 based on the same test system, but with more test cases. The same IDS is applied to this dataset using the DT-CWT and AlexNet, with identical parameters. The proposed XAI methodology using RISE was applied to the results, and explanations generated for each case in the new dataset. The quantitative metrics are calculated for RISE for the new validation dataset and compared with those of the original testing dataset in Table 17. As a newly trained CNN on new data is used, despite being from the same system, there is still some variation in the metrics due to the randomness in the process of training a CNN. Compared to the results from testing different XAI methodologies on the testing dataset, the changes seen in RISE moving from the testing dataset to the new verification dataset are not significant, and the values are in line with RISE. This shows that the quality of the explanations has remained stable with a new dataset and that the results in this work are generalizable to larger datasets.

5.6. Limitations

This section addresses the major limitations of this work, including this work's focus on image-based data, the time-frequency spectrograms used, and the dependence of the selected parameters to the models and data used.

Table 16
A breakdown of the verification dataset used in this work.

Class	Number of Cases
Normal	635
Disturbance	179
Attack, FDI	687
Attack, Replay	89
Attack, MS	276
Attack, DoS	806
Attack, Insider	512
Total Cases	**3184**

Table 17
A comparison between the original testing dataset and the verification dataset, metrics are averaged across the entire dataset.

Dataset	Sparsity	TV	Stability	AUC
Testing	0.0852	0.0087	0.0378	0.1409
Verification	0.0989	0.0097	0.0437	0.1364

5.6.1. Focus on image-based data

This work focuses on specifically time-frequency spectrograms, which are a case of images. The given methodology and analysis still hold on general images, as the same analysis can be used. This is as it was desired in this work to provide a visual analysis based on XAI methods not already covered in literature. In the case of non-image-based data, such as numerical vector-formatted data, the given methods do not all apply directly.

While LIME and SHAP have implementations that are suited for other data types and can be used, VG, SmoothGrad, and Grad-CAM as outlined in this work can only be applied if the data is preprocessed into images. OSM and RISE meanwhile can be adapted specifically to the input data, and care must be taken to ensure that the resulting explanations are meaningful. Alternatively, the non-image data types can be converted into images by organizing them and transforming them into a time-frequency spectrogram. This is the processing technique used in this work to convert the collected IEC 61850 GOOSE data into spectrograms.

5.6.2. Dependence of parameter selection and results to specific SDS and IDS architecture

While this work focuses on a specific experimental SDS cybersecurity dataset and application, alternate datasets and different model architectures can change the specific parameter values required for the best results. As such, this work does not try to provide a specific parameter value across all datasets. The numbers provided in this work are relevant to the system investigated. The process used in this work should be used to determine the optimal parameters for the desired explanations for each unique SDS system and IDS architecture. In this case, the guidance provided still holds and is relevant.

5.7. Implications on regulatory, safety, and operational implications

The XAI techniques investigated make ML-based and DL-based IDSs compliant with the European Union's General Data Protection Regulation. The models, whose decisions can negatively impact utilities as well as the broad consumers of electricity as the decisions of the models are now explainable and satisfy the need to make decisions that can affect individuals negatively [1]. Additionally, the explainability offered by XAI provides a way to evaluate trust and allows for improved comprehensive regulations regarding AI to mitigate AI's risk as they are

currently limited in by their inability to evaluate trust [36].

Improved safety is achieved as when classifications are made and decisions taken, reasons are provided that can lead operators and engineers to the root cause of a scenario, facilitating faster and more effective solutions. The explanations can also identify what features led to a disturbance for example and allow engineers to fix the root cause with less trial and error reducing further potential damages and harm.

In an operational environment the use of XAI leads to additional system complexities. The generation of explanations adds complexity to an IDS as how explanations are generated, displayed, and/or stored must be considered. Additional hardware may be required to generate these explanations in a timely fashion, such as more powerful GPUs capable of accelerating DL models.

In an actual SDS IDS operational environment, the input data from the SDS collected is known to be from specific sources. Once an AI model is trained, by using XAI to generate explanations operators and engineers can analyze the explanations and draw links between each explanation and the important features and the physical implications of such features, like what is done in Section 3 and Fig. 5. Once causal explanations of what features led to what attacks are drawn it can be verified if the chosen AI has effectively learned such features and is not overfitted or has learned spurious features.

This reassures operators that the decisions made by the model are grounded in reality and can be used as a basis for further decision-making. The explanations increase the operators' trust in the SDS IDS and leads the way to improved adoption of AI for IDS in this context.

6. Conclusion

This work has provided an experimental review of several XAI techniques, including those popular in the literature, from the perspective of integrating XAI into IDS for cybersecurity in SDS. Using data from a SDS cybersecurity test system, the identified XAI techniques have been investigated in terms of their parameters, human interpretability, stability, trustworthiness, and fairness. The key parameters of each identified XAI technique have been investigated experimentally with a sensitivity analysis, and guidance on how to select the parameters to improve explanations are highlighted.

A summary of the investigated XAI techniques has been provided, and the key benefits and drawbacks of each XAI technique have been identified. Overall, issues of variability, low-resolution explanations, inaccurate explanations, and less meaningful explanations can be addressed by carefully selecting an XAI methodology that does not suffer from the key issue at hand. However, it appears that computational complexity and overall quality of explanations is a trade-off and should be taken into consideration when selecting an XAI methodology. For the given experimental test system and IDS, the results have shown that qualitatively, for the highest resolution, meaningful, and stable explanations, RISE should be used. Through its random sampling, it can best gather the model's behaviour across the input space for a test case. An analysis of the parameters of RISE showed that generally, selecting a balance is best, and that for the number of samples there is a balanced point where quality explanations are guaranteed, and computational complexities can be reduced. However, evaluation of the behaviour should be conducted for any new test system. Among the techniques, generally as the quality and detail of the explanations increase, so does the computational complexity of the methods. Quantitative analysis of the explanations of each identified XAI method has been provided and should be used as guidance in tandem with the qualitative comparisons provided. This work can be used as reference in the development of the next generation of XAI-enhanced SDS and SG applications, as it can be used to help guide the selection of an XAI technique to be integrated with IDS, and how to better select the parameters of such technique. By facilitating the adoption of XAI techniques and a more informed application of such techniques, this work serves to advance the adoption of XAI in the field of SDS cybersecurity to increase AI models' trustworthiness.

CRediT authorship contribution statement

M. Oinonen: Writing – original draft, Methodology, Investigation, Formal analysis. **W.G. Morsi:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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